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RESEARCH ARTICLE

Comprehensive Benchmarking of Attention Enhanced CNNs and Transfer Learning Models for Real Time Driver Drowsiness Detection

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ABSTRACT Driver drowsiness is a leading cause of traffic accidents worldwide, highlighting the urgent need for accurate and real-time detection systems to enhance road safety. Recent advances in deep learning have shown great promise, particularly with attention-augmented convolutional neural networks (CNNs). However, most existing studies focus on a single attention mechanism in isolation, providing limited insight into their comparative effectiveness. In this work, we present a systematic and comprehensive framework for integrating, analyzing, and benchmarking multiple attention mechanisms within a custom CNN architecture tailored for driver drowsiness detection. Specifically, we investigate four representative modules: Squeeze-and-Excitation (SE) channel attention, Spatial Attention, Efficient Channel Attention (ECA), and the hybrid Convolutional Block Attention Module (CBAM). A baseline CNN without attention is implemented to enable direct assessment of the added value of each mechanism. Furthermore, we benchmark our framework against six widely adopted transfer learning models, including VGG19, DenseNet169, ResNet50V2, InceptionV3, MobileNet, and InceptionResNetV2. Extensive experiments conducted on the NTHU-DDD2 dataset under real-world driving conditions demonstrate that attention mechanisms significantly enhance classification performance. Among them, CBAM achieved the best trade-off between accuracy and efficiency, reaching 99.63% accuracy, an AUC of 0.9999, and competitive inference latency suitable for real-time deployment. ECA also performed strongly, validating the role of lightweight attention in improving recall and training stability. By providing the first comprehensive comparative study of diverse attention mechanisms in this domain, this work establishes a robust benchmark and demonstrates that hybrid attention-augmented CNNs, particularly CBAM, are highly effective for driver drowsiness detection. These findings advance the development of intelligent and safety-critical transportation systems, while outlining pathways for future research toward robust, efficient, and deployable solutions.

INDEX TERMS Driver drowsiness detection, attention mechanisms, convolutional neural networks (CNNs), convolutional block attention module (CBAM), efficient channel attention (ECA), squeeze-and-excitation (SE), spatial attention, transfer learning, intelligent transportation systems (ITS).

I. INTRODUCTION

Driver drowsiness refers to a state of fatigue or sleepiness experienced by the driver, which substantially impairs the

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ability to operate a vehicle safely [1]. This condition negatively affects decision-making capacity, reaction time, and situational awareness, thereby constituting one of the major contributors to severe and often fatal traffic accidents. Empirical evidence has shown that driving under drowsy conditions can be as detrimental as driving under the influence of alcohol, as both impairments reduce the driver's ability to maintain lane position, accurately evaluate distances, and effectively process visual and auditory cues from the driving environment [2]. The implications of driver drowsiness are therefore multifaceted, extending beyond the immediate threat to individual drivers to broader concerns of public health, economic stability, and road safety.

According to numerous studies, fatigue-related drowsiness is responsible for approximately 20% to 50% of road accidents worldwide [3], [4]. The consequences of such accidents manifest not only in terms of human casualties but also in significant financial burdens. From an economic perspective, drowsiness-induced accidents result in considerable expenditures related to vehicle repair, healthcare services, insurance premiums, and legal liabilities. For instance, the U.S. National Highway Traffic Safety Administration (NHTSA) has estimated that fatigue-related crashes impose an economic loss of approximately $12.4 billion annually [5]. On a personal level, the risk is even more alarming, as drowsy driving significantly elevates the likelihood of injuries and fatalities. Road safety statistics report that fatigue contributes to nearly 37% of the 1.35 million annual traffic-related deaths, making it a leading cause of mortality among individuals aged five to 29 and ranking as the ninth most common cause of death globally [6].

Beyond the direct health and economic implications, driver drowsiness exerts a broader societal impact by contributing to traffic congestion, emergency response delays, and post-accident cleanup operations. These disruptions further exacerbate the loss of economic productivity and reduce the overall efficiency of transportation systems [7], [8]. Consequently, addressing the challenge of driver drowsiness is not merely a matter of individual safety but a critical component of ensuring the sustainability, resilience, and reliability of modern transportation networks.

Recent advances in computer vision and deep learning have revolutionized the field of object recognition, enabling machines to automatically detect and classify objects in images and video streams with unprecedented accuracy. Within this broader domain, driver monitoring systems, particularly those designed for fatigue and drowsiness detection, have emerged as an area of critical importance, given the potentially catastrophic consequences of undetected driver impairment. Unlike conventional object detection tasks, where targets are typically static or exhibit consistent structural features, drowsiness detection involves recognizing subtle, transient, and non-stationary behavioral cues. These include micro-expressions such as eye blinks, partial eyelid closures, head tilts, gaze deviations, and mouth movements associated with yawning.

The dynamic and context-dependent nature of these indicators renders traditional object detection approaches inadequate, as they fail to capture temporal dependencies and fine-grained variations in driver behavior. Consequently, more advanced methodologies are required that not only incorporate spatial feature extraction but also model temporal dynamics over consecutive frames. This necessitates the design of specialized architectures that combine the strengths of deep convolutional models with temporal feature modeling, thereby enabling the reliable identification of early warning signs of fatigue. Addressing this challenge is essential for the development of intelligent, real-time monitoring systems capable of preventing accidents and enhancing road safety in complex and unpredictable driving environments.

Convolutional Neural Networks (CNNs) have emerged as the cornerstone architecture for modern computer vision tasks due to their exceptional ability to capture spatial hierarchies and extract multi-level feature representations from image data [9], [10], [11]. In the context of driver drowsiness detection, CNNs are widely applied to analyze sequential video frames or still images in order to monitor behavioral cues such as variations in facial expressions, prolonged eye closures, gaze patterns, and yawning frequency. The effectiveness of these models depends critically on their capacity to process high-dimensional visual inputs while isolating the subtle features that serve as early indicators of fatigue.

State-of-the-art CNN architectures, including ResNet, Inception, and DenseNet, have demonstrated remarkable performance in related detection tasks, owing to their depth, modularity, and ability to capture both low-level and high-level abstractions. These architectures leverage residual connections, inception modules, and dense connectivity to mitigate issues such as vanishing gradients and feature redundancy, thereby enabling robust learning of intricate visual cues relevant to driver monitoring. Furthermore, transfer learning has significantly enhanced the practical applicability of CNN-based systems by fine-tuning pre-trained models on domain-specific datasets. This approach not only accelerates the training process but also improves generalization, making it particularly valuable for driver drowsiness detection, where the availability of large-scale annotated datasets is often limited. As a result, CNNs combined with transfer learning constitute a powerful paradigm for developing accurate, efficient, and scalable fatigue detection systems.

Despite the remarkable success of CNNs and transfer learning architectures, their capacity to capture the fine-grained and transient variations in facial dynamics remains limited. Critical yet subtle indicators of drowsiness, such as partial eyelid closures, fleeting yawns, micro head tilts, or gaze shifts, can easily be underrepresented in conventional feature extraction pipelines. When these minor but highly discriminative cues are overlooked, the reliability of drowsiness detection systems is significantly reduced, particularly in real-world driving conditions where illumination, occlusion, and motion artifacts further complicate recognition.

Research Problem: Although CNNs and transfer learning architectures have shown strong potential in driver drowsiness detection, their performance is often constrained by the inability to reliably capture fine-grained, transient facial cues that signify early stages of fatigue. Subtle indicators such as micro-expressions, partial eyelid closures, and brief yawns are frequently underrepresented in conventional feature extraction pipelines, leading to reduced robustness under real-world conditions with varying illumination, occlusions, and motion artifacts. This limitation underscores the need for more advanced architectures that can effectively emphasize discriminative features while suppressing irrelevant noise.

Research Motivation: The increasing number of accidents attributed to driver fatigue highlights the urgent demand for intelligent monitoring systems capable of real-time detection. Developing models that not only achieve high accuracy but also maintain computational efficiency is crucial for deployment in vehicles, where hardware resources may be constrained. Moreover, while attention mechanisms have demonstrated improvements in various computer vision tasks, their systematic application and comparative evaluation in the context of driver drowsiness detection remain underexplored. This research is motivated by the need to fill this gap and establish a benchmark framework that bridges theoretical advancements with practical safety-critical applications in intelligent transportation systems (ITS).

To address this limitation, attention mechanisms have emerged as a powerful complement to CNN architectures, enabling networks to adaptively prioritize the most informative spatial regions or channel-specific features. By dynamically recalibrating feature maps, attention modules guide the network to focus on regions of interest, suppress irrelevant background noise, and enhance the discriminability of subtle facial cues. Multiple variants of attention have been proposed, including Squeeze-and-Excitation (SE) channel attention, which emphasizes inter-channel dependencies; Spatial Attention, which highlights critical regions within feature maps; Efficient Channel Attention (ECA), which reduces computational complexity while preserving performance; and hybrid modules such as the Convolutional Block Attention Module (CBAM), which jointly integrates spatial and channel-level attention.

These mechanisms enrich the representational power of CNNs, offering a more nuanced understanding of facial and behavioral cues associated with driver fatigue. As such, the incorporation of attention into CNN-based drowsiness detection frameworks represents a significant step toward developing robust, accurate, and real-time intelligent driver monitoring systems.

In this study, we propose a comprehensive deep learning framework for driver drowsiness detection that systematically explores and evaluates the impact of diverse attention mechanisms within a custom-designed CNN architecture. To establish reliable feature extraction, DLib is first employed to accurately localize key facial landmarks, including the face, left eye, right eye, and mouth, which serve as the primary regions of interest for fatigue assessment. Unlike prior research that predominantly investigates a single attention strategy in isolation, our framework provides a comparative analysis by juxtaposing a baseline CNN against four attention-augmented variants: SE, Spatial Attention, ECA, and the CBAM. This design enables a systematic assessment of the relative contributions of channel-only, spatial-only, lightweight, and hybrid attention mechanisms in enhancing drowsiness classification.

Beyond this core comparison, we extend our analysis by benchmarking the attention-enhanced CNNs against six widely recognized transfer learning models, VGG19, DenseNet169, ResNet50V2, InceptionV3, InceptionResNetV2, and MobileNet. This dual-layered evaluation positions our findings within the broader landscape of state-of-the-art architectures, allowing us to highlight both the strengths and limitations of attention integration relative to well-established deep learning baselines. By offering the first holistic comparison across multiple attention mechanisms in this domain, our work provides an essential benchmark for future research while advancing the development of intelligent, accurate, and real-time driver monitoring systems.

Extensive experiments conducted on real-world driving datasets demonstrate that incorporating attention mechanisms into CNN architectures significantly enhances both the accuracy and robustness of drowsiness classification compared to the baseline CNN. Among the evaluated variants, the CBAM consistently achieves the most favorable balance across multiple performance metrics, including accuracy, precision, recall, and F1-score, surpassing 99% accuracy while maintaining stable reliability. Beyond predictive performance, we further assess computational efficiency to underscore the practicality of deploying these models in real-world driving environments, where low latency and resource constraints are critical.

The experimental findings validate the effectiveness of attention integration for fatigue detection and establish CBAM-CNN as a particularly promising candidate for real-time vehicular monitoring systems. By bridging the gap between accuracy and efficiency, this study not only provides empirical evidence of the advantages of hybrid attention mechanisms but also contributes a benchmark framework that can guide future research and development of intelligent transportation systems aimed at mitigating fatigue-related accidents.

Research Contribution: The primary contributions of this work can be summarized as follows:

- We propose a systematic comparative framework that integrates and evaluates multiple attention mechanisms, namely SE, Spatial Attention, ECA, and the CBAM, within a custom CNN architecture for driver drowsiness detection.

- A baseline CNN model is established to serve as a reference point, enabling the quantitative assessment of performance improvements introduced by each attention mechanism across multiple metrics, including accuracy, precision, recall, and F1-score.
- The proposed attention-augmented CNN models are comprehensively benchmarked against six widely adopted state-of-the-art transfer learning architectures (VGG19, DenseNet169, ResNet50V2, InceptionV3, InceptionResNetV2, and MobileNet), thereby situating our findings within the broader landscape of deep learning solutions for fatigue detection.
- Experimental evaluations demonstrate that the CBAM-enhanced CNN achieves superior classification performance while maintaining computational efficiency, making it well-suited for real-time vehicular deployment under practical resource constraints.
- To the best of our knowledge, this work presents the first comprehensive benchmark study of attention mechanisms for driver drowsiness detection, providing critical insights and a robust evaluation framework that can inform future research and guide the development of ITS systems.

Research Objectives: Building upon the identified challenges and motivations, this work aims to achieve the following objectives:

- To design and implement a custom CNN-based framework that systematically integrates multiple attention mechanisms for driver drowsiness detection.
- To conduct a comparative analysis of different attention strategies (SE, Spatial, ECA, CBAM) in order to evaluate their relative effectiveness in enhancing feature representation.
- To benchmark the proposed attention-augmented CNNs against state-of-the-art transfer learning architectures and assess their performance in terms of accuracy, robustness, and computational efficiency.
- To establish a reference benchmark that can guide future research on attention-based models for fatigue detection within the broader domain of ITS.

The remainder of this paper is organized as follows. Section II provides a comprehensive review of related works on driver drowsiness detection, highlighting both traditional and deep learning-based approaches with emphasis on attention mechanisms. Section III introduces the proposed framework, detailing the custom CNN architecture, the integration of four distinct attention mechanisms (SE, Spatial, ECA, and CBAM), and the transfer learning baselines considered for comparison. Section IV presents the experimental setup, dataset distribution, training configuration, and evaluation metrics, followed by an in-depth analysis of the results across all models. Section V benchmarks the proposed approach against state-of-the-art methods reported in the literature to contextualize its performance. Section VI analyzes the findings, highlighting key observations regarding accuracy, computational efficiency, training stability, and latency trade-offs for real-time deployment. Finally, Section VII summarizes the main contributions, discusses limitations, and outlines avenues for future research to enhance the robustness and applicability of driver drowsiness detection systems.

II. LITERATURE REVIEW

CNNs have been widely adopted as the foundational architecture for driver drowsiness detection due to their ability to learn hierarchical spatial features from image data. In [12], the authors employed a CNN-based framework to detect drowsiness by leveraging DLib for face localization and subsequently computing the Mouth Aspect Ratio (MAR) as a key discriminative feature. To address the limitations posed by a relatively small dataset, data augmentation techniques were applied, enabling the categorization of driver states into ''yawning'' and ''no_yawning.'' The study reported an accuracy of 96.69% on the unaugmented dataset, demonstrating the potential of CNNs in capturing fatigue-related features. However, while effective for binary classification of yawning behaviors, the reliance on a single handcrafted feature (MAR) restricts the generalizability of the approach, particularly in detecting more subtle or diverse indicators of driver fatigue such as eye closure duration or head movement patterns. This highlights the need for more comprehensive architectures capable of integrating multiple feature representations to achieve robust real-world applicability.

In [13], the authors introduced a vision-based driver monitoring system called *DriCar*, which employs a CNN model to predict driver exhaustion by analyzing behavioral cues such as yawning frequency, eye blinks, and eyelid closure duration. Unlike approaches that rely on intrusive physiological sensors or wearable devices, DriCar leverages in-vehicle video recording to unobtrusively capture facial dynamics, thereby improving driver comfort and system usability. The deep learning model processes video streams in real time and achieves an accuracy rate of approximately 92%, demonstrating the feasibility of camera-based fatigue detection. However, while effective in controlled environments, this method remains vulnerable to external challenges such as variable lighting conditions, head pose variations, and partial occlusions (e.g., sunglasses or masks). These limitations suggest that although DriCar represents an important step toward practical non-intrusive monitoring, further enhancements, particularly in feature robustness and adaptability, are required for reliable deployment in diverse real-world driving scenarios.

In [14], the authors proposed a hybrid methodology that integrates both invasive and non-intrusive approaches for drowsiness detection. Specifically, skin conductance levels were measured using a Galvanic Skin Response (GSR) sensor to capture physiological variations associated with fatigue, while facial features were extracted through Multi-Task Cascaded Convolutional Neural Networks (MTCNN) to model behavioral cues. The combined framework was evaluated on the NTHU-DDD dataset, which includes recordings from 36 subjects exhibiting varying levels of sleepiness, and achieved an overall accuracy of 91%. This study highlights the potential benefits of multimodal systems that fuse physiological and visual indicators to enhance detection reliability. Nevertheless, the reliance on invasive hardware such as GSR sensors may reduce user comfort and hinder scalability for large-scale vehicular deployment. These drawbacks underscore the importance of advancing non-intrusive, vision-based solutions that balance accuracy with practicality in real-world driving environments.

In [15], the authors proposed a hybrid deep learning framework that integrates CNNs with Long Short-Term Memory (LSTM) networks to enhance driver sleepiness prediction. While the CNN component is responsible for extracting spatial features from facial images, the LSTM module captures temporal dependencies across consecutive frames, thereby enabling the model to recognize fatigue-related behavioral patterns that evolve over time. The framework was trained and validated on the MRL Eye Dataset and the NTHU-DDD dataset, demonstrating superior accuracy compared to several state-of-the-art techniques. This work underscores the effectiveness of combining spatial and temporal modeling for drowsiness detection. However, the increased architectural complexity and higher computational overhead introduced by the CNN–LSTM hybrid may present challenges for real-time vehicular deployment, particularly in resource-constrained environments.

In [16], the authors investigated the effectiveness of facial landmarks and handcrafted features for assessing driver tiredness. The Video-Based Driver Drowsiness Detection (VBDDD) dataset was employed, and three classifiers, Random Forest, Sequential Neural Network, and Support Vector Machine (SVM), were applied to perform the classification task. To improve feature discriminability, the Common Spatial Pattern (CSP) method was adopted, resulting in an accuracy of 94% and improved class sample balance. While this study demonstrates that classical machine learning methods, combined with feature engineering, can achieve competitive performance, such approaches remain highly dependent on manual feature selection and are less adaptive to variations in real-world driving conditions. Compared to end-to-end deep learning frameworks, they may lack the scalability and robustness required for capturing subtle and dynamic facial cues across diverse environments.

In [17], the authors utilized a Kaggle-based drowsiness dataset consisting of 2,900 images categorized into four classes: yawning, not yawning, eyes open, and eyes closed. Deep learning models were trained to detect drowsiness-related eye movements, with both a custom CNN and the pre-trained VGG16 architecture being evaluated. The results demonstrated that the CNN model outperformed VGG16, achieving 97% accuracy and a 99% F1-score, underscoring the effectiveness of lightweight, task-specific models in this domain. While the study highlights the potential of deep learning for robust eye-state classification, its reliance on a relatively small dataset poses challenges for generalization, as performance may degrade when exposed to diverse driving conditions and varied subject characteristics. This limitation reinforces the need for larger, more representative datasets and advanced architectures capable of capturing fine-grained, real-world variability.

In [18], the authors proposed a Hierarchical Deep Drowsiness Detection (HDDD) network that combines both spatial and temporal modeling for fatigue recognition. The framework employs ResNet in the initial stage to extract spatial features, enabling accurate identification of critical facial attributes such as illumination conditions, presence of eyewear, and detailed localization of the eyes and lips. This stage significantly improves the precision of facial feature extraction under varying environmental conditions. In the subsequent stage, a LSTM network is utilized to capture temporal dependencies across video frames, thereby modeling the sequential evolution of drowsiness-related behaviors. The HDDD framework achieved an accuracy of 87.1%, illustrating the potential of hierarchical architectures that fuse spatial and temporal deep networks. However, despite its conceptual strength, the relatively modest accuracy highlights the need for further optimization, particularly in balancing complexity with robustness for real-world applications.

In [19], the authors proposed a hybrid deep learning approach that integrates CNNs with Bidirectional Long Short-Term Memory (BiLSTM) networks to improve driver drowsiness detection. The system leverages a video camera to continuously monitor blinks and facial movements, operating in three key stages: (i) webcam-based face detection, (ii) ocular feature extraction using the Euclidean distance method, and (iii) real-time eye-blink monitoring to identify signs of fatigue. By combining CNNs for spatial feature extraction with BiLSTMs for capturing bidirectional temporal dependencies, the model achieved an accuracy of 94%. While this approach demonstrates the effectiveness of spatio-temporal modeling for fatigue detection, its reliance on handcrafted ocular feature extraction introduces potential limitations, as performance may degrade under conditions such as partial occlusion, head pose variation, or inconsistent lighting. These challenges highlight the importance of fully end-to-end deep learning architectures that can robustly generalize across diverse real-world scenarios.

In [20], the authors developed a sleepiness detection framework that utilized the NTHU-DDD dataset and achieved an accuracy of 85.62%. Their approach combined an advanced Histogram of Oriented Gradients (HOG) feature extractor with a Naive Bayes classifier to distinguish between drowsy and non-drowsy states. While the use of handcrafted HOG features proved effective for capturing gradient-based facial patterns, the overall performance lagged behind contemporary deep learning approaches. More importantly, the model exhibited poor generalizability, as handcrafted feature extractors and shallow classifiers are often highly dataset-dependent and prone to performance degradation when applied to heterogeneous real-world driving conditions. This underscores the necessity of adopting end-to-end deep learning frameworks that can automatically learn robust, transferable representations for fatigue detection across diverse environments.

In [21], the authors proposed a two-stage vision-based framework for real-time driver fatigue detection utilizing Facial Action Units (FAUs), which capture fine-grained facial movements such as eye widening and jaw dropping. In the first stage, a CNN was employed to detect FAUs, while in the second stage, an Extreme Gradient Boosting (XGBoost) classifier was used to determine drowsiness levels. The system achieved a notable accuracy of 96% on the NTHU-DDD dataset, demonstrating the effectiveness of combining deep learning for feature extraction with ensemble learning for classification. Nevertheless, the framework's real-time applicability remains limited, as its performance and efficiency are constrained when scaled to larger, more diverse datasets. This highlights the broader challenge of ensuring that high-performing fatigue detection models also maintain robustness and scalability under real-world conditions.

In [22], the authors proposed an ensemble deep learning framework for driver drowsiness detection that integrates both mouth and eye features. The model architecture consists of two InceptionV3 modules designed to maintain a compact parameter space while effectively capturing discriminative visual cues. Feature extraction is performed using MTCNN on facial subregions, after which a weighted average ensemble strategy, optimized through an ensemble tuning algorithm, combines the outputs to determine the driver's drowsiness state. Evaluated on the NTHU-DDD dataset, the framework achieved an accuracy of 97.1%, demonstrating robustness against variations in posture and illumination. Although the ensemble strategy enhances resilience and classification performance, the reliance on multiple deep learning modules increases computational demands, which may hinder deployment in resource-constrained vehicular environments. This trade-off highlights the continuing need for architectures that balance accuracy with efficiency for real-time monitoring applications.

In [23], the authors introduced a novel isotropic self-supervised learning (IsoSSL) framework designed to construct powerful image representations without requiring manual annotations. Building upon this foundation, they further proposed a hybrid extension termed IsoSSL-Momentum Contrast (IsoSSL-MoCo). To fully exploit the complementarity of multimodal information, the study also presented an attention-based fusion model that integrates features extracted from the lips, eyes, and optical flow of head movements. The image encoders for these modalities were first pre-trained using the IsoSSL strategy and subsequently fine-tuned within the fusion network. Experimental results demonstrated strong performance, with the proposed approach achieving accuracies of 93.71% on the NTHU-DDD dataset and 98.65% on the YawDD dataset. While this work highlights the promise of self-supervised pre-training and multimodal fusion in fatigue detection, its reliance on multiple data streams and pre-training complexity may limit real-time applicability in vehicular settings, where lightweight and computationally efficient models are required.

In [24], the authors proposed a Joint Head Pose and Facial Action Network (JHPFA-Net) for robust yawning detection across varying head positions in video sequences. The architecture is composed of three key modules: (i) a dual-channel Head Pose and Facial Action Fusion Module (HF-Module), which integrates head pose information with facial features; (ii) a Face Frontalization with Warp Attention Module (FF-Module), which generates synthetic frontal facial representations to preserve expression details under pose variations; and (iii) a Geometric-based Keyframe Selection Module (GK-Module), which selects the most informative frames for efficient analysis. Experimental results on the YawDD dataset reported an accuracy of 86.7%, demonstrating the potential of pose-aware learning for fatigue-related facial expression analysis. Nonetheless, while JHPFA-Net effectively addresses challenges related to head orientation, its overall performance remains comparatively modest, suggesting that further improvements in feature representation and attention modeling are necessary to achieve higher robustness in real-world driving scenarios.

In [25], the authors proposed a Transfer Learning Population-Based Sampling Network (TLPSN) aimed at improving the generalizability of drowsiness detection models across different viewing angles. The framework leverages transfer learning to adapt pre-trained models, while population-based sampling techniques are employed to balance data distribution and reduce bias during training. A driver-in-the-loop platform was utilized to collect a new dataset for evaluation, enabling the system to capture realistic variations in driver behavior. Experimental results demonstrated a high accuracy of 97.5% on a private dataset, highlighting the effectiveness of transfer learning combined with advanced sampling strategies for enhancing robustness to pose variability. Nevertheless, the reliance on a private dataset raises concerns regarding reproducibility and benchmarking, as direct comparisons with widely used public datasets remain limited. This restricts the broader validation and scalability of the approach within the research community.

In [26], the authors developed two complementary models for driver drowsiness detection, designed to separately capture temporal and spatiotemporal features. Model-A focused on temporal dynamics by first extracting features using YOLOv3 and subsequently processing them through an LSTM network. To mitigate overfitting, TransGAN-based data augmentation was employed, enhancing the variability of the training samples. In contrast, Model-B concentrated on spatiotemporal representations, leveraging transfer learning with the InceptionV3 architecture to extract spatial features while simultaneously modeling temporal dependencies. Comparative evaluations demonstrated that Model-B outperformed Model-A, achieving an accuracy of 97.5% compared to 86% for Model-A. This study underscores the importance of incorporating spatiotemporal modeling for fatigue detection. However, the reliance on complex multi-stage pipelines increases computational demands, potentially limiting real-time applicability in vehicular settings where efficiency and low latency are critical.

In [27], the authors conducted a comparative study of seven transfer learning architectures, VGG19, MobileNetV2, InceptionResNetV2, InceptionV3, Xception, ResNet50V2, and DenseNet169, using the NTHU-DDD2 dataset. The dataset was partitioned into 70% for training, 20% for validation, and 10% for testing to ensure a robust evaluation protocol. Among the evaluated models, VGG19 achieved the highest performance following extensive hyperparameter optimization, with an accuracy of 96.51%, an F1-score of 96.73%, a precision of 98.14%, and a recall of 95.36%. This study underscores the effectiveness of transfer learning for fatigue detection, particularly when model tuning is carefully performed. However, while transfer learning offers strong performance, it often comes at the cost of increased computational overhead, making deployment challenging in real-time vehicular environments. This limitation reinforces the need for approaches that balance predictive accuracy with computational efficiency.

In [28], the authors proposed a hybrid deep learning framework that integrates LSTM networks with multi-granularity facial features for drowsiness detection. A Practical Facial Landmark Detector (PFLD) was employed to identify key facial landmarks, enabling the extraction of localized features such as eye and mouth regions. In parallel, a Vision Transformer (ViT) model was utilized to capture higher-level semantic and contextual representations of driver behavior. The extracted features were fused and processed through the LSTM to model temporal dependencies across video frames. Experimental evaluations on the NTHU-DDD dataset demonstrated an accuracy of 93.15%, validating the effectiveness of combining local feature detection with transformer-based semantic modeling. However, despite its promising results, the increased complexity of incorporating multiple deep learning components raises concerns regarding real-time feasibility, especially in vehicular environments with limited computational resources.

Table 1 provides a comparative overview of existing studies on driver drowsiness detection. It is evident that while classical machine learning techniques (e.g., HOG + Naive Bayes, CSP + SVM) achieved only moderate performance, they fall short of modern deep learning frameworks in terms of accuracy and robustness. Hybrid deep learning approaches, such as CNN–LSTM, BiLSTM, and ensemble-based models, demonstrated stronger predictive power by leveraging both spatial and temporal dependencies. Transfer learning architectures, particularly VGG19 and Inception-based models, further improved performance, often exceeding 96% accuracy. More recent multimodal and self-supervised methods showed promise in addressing posture variation and data scarcity. Nonetheless, despite these advances, the proposed CBAM-augmented CNN framework consistently outperforms prior approaches, achieving over 99% accuracy while maintaining computational efficiency suitable for real-time deployment. This establishes our method as a strong benchmark for fatigue detection in intelligent transportation systems.

Despite these advances, several limitations remain pervasive across the literature. First, many methods rely heavily on handcrafted features (e.g., MAR, HOG, CSP), which are highly sensitive to illumination changes, head pose variation, and occlusion, thereby limiting generalizability. Second, although hybrid CNN–LSTM and BiLSTM architectures effectively capture temporal dynamics, their increased complexity incurs significant computational costs, reducing suitability for real-time vehicular applications. Third, transfer learning models, while powerful, introduce considerable overhead and often struggle to adapt to smaller, domain-specific datasets. Finally, multimodal and self-supervised approaches improve robustness but require additional modalities, sensors, or extensive pre-training pipelines, complicating deployment in practical driving environments. To address these gaps, the proposed work introduces a unified CNN framework enhanced with diverse attention mechanisms, SE, ECA, Spatial Attention, and the hybrid CBAM. By adaptively emphasizing salient visual cues while suppressing redundant information, these attention modules improve robustness to environmental variability and achieve superior classification accuracy. Crucially, this design balances predictive performance with computational efficiency, positioning the proposed CBAM-CNN as a scalable and practical solution for real-time driver drowsiness detection.

Building on these insights, the following section introduces the proposed methodology, detailing the architectural design, the integration of attention mechanisms, and the benchmarking strategy against state-of-the-art models.

III. PROPOSED METHODOLOGY

The proposed methodology is designed to construct a robust deep learning framework for driver drowsiness detection, with the objective of accurately distinguishing between the "drowsy" and "not drowsy" states from visual data. The framework is developed to ensure both high classification accuracy and computational efficiency, making it suitable for real-time deployment in intelligent transportation systems. The overall pipeline begins with the acquisition of input images, which are processed into sequential frames to capture temporal continuity within the driving scenario. To precisely isolate the most informative regions, facial landmark detection is performed using the DLib library, which has

TABLE 1. Summary of the reviewed literature and comparison with the proposed work.

Ref	Year	Dataset	Techniques	Accuracy
[13]	2019	CelebA, YawDD, volunteer video data	CNN (DriCar System)	92%
[18]	2021	NTHU-DDD	Hierarchical Deep Network (ResNet + LSTM)	87.19%
[19]	2021	Eye Image Dataset	CNN + BiLSTM (blink monitoring)	94%
[20]	2021	NTHU-DDD	HOG Features + Naive Bayes Classifier	85.62%
[21]	2021	NTHU-DDD	CNN + XGBoost (FAUs)	96%
[22]	2021	NTHU-DDD	Dual InceptionV3 + MTCNN (Ensemble)	97.1%
[23]	2021	NTHU-DDD, YawDD	IsoSSL-MoCo + Attention Fusion	93.71%, 98.65%
[15]	2022	NTHU-DDD, MRL Eye	CNN + LSTM Hybrid	98%
[25]	2022	Private Dataset	TLPSN + MTCNN (Transfer Learning)	97.5%
[14]	2023	NTHU-DDD	Hybrid (GSR + MTCNN)	91%
[16]	2023	VBDDD	Random Forest, SNN, SVM + CSP Features	94%
[17]	2023	Kaggle Drowsiness Dataset	CNN vs. VGG16 (CNN Superior)	97%
[12]	2023	YawDD	CNN + MAR (yawn detection)	96.69%
[24]	2023	YawDD	JHPFA-Net (HF, FF, GK Modules)	86.7%
[26]	2023	UTA-RLDD	Model-A (YOLOv3+LSTM), Model-B (InceptionV3)	86%, 97.5%
[28]	2024	NTHU-DDD	Multi-Granularity Features + ViT + LSTM	93.15%
[27]	2024	NTHU-DDD2	Transfer Learning (VGG19 best)	96.51%
Proposed Work	2026	NTHU-DDD, Real-World Driving Data	CNN + Attention Mechanisms (SE, ECA, Spatial, CBAM)	**99%+ (CBAM)**

proven effective in reliable facial alignment tasks. Through this process, critical facial regions such as the mouth, face, right eye, and left eye are accurately localized, providing the foundation for feature extraction. These localized regions are essential since micro-expressions, including partial eye closures, yawning, and subtle head movements, serve as primary indicators of drowsiness. By focusing on these regions, the framework ensures that the subsequent feature extraction process emphasizes discriminative visual cues while minimizing redundant background information.

Building on the extracted facial regions, a custom CNN architecture is developed to serve as the backbone of the proposed framework. To enhance the discriminative capability of the CNN, we systematically integrate and evaluate multiple attention mechanisms, thereby enabling the network to adaptively prioritize informative features while suppressing redundant or irrelevant details. Attention mechanisms are particularly crucial in the context of driver drowsiness detection, where subtle cues such as partial eyelid closures, yawning, or slight head tilts must be distinguished from normal facial expressions under varying environmental conditions. To this end, four distinct strategies are investigated within our framework:

- SE Channel Attention: enhances representational power by adaptively recalibrating channel-wise feature responses, allowing the network to emphasize the most relevant feature channels.
- Spatial Attention: highlights critical spatial locations within feature maps, ensuring that localized facial cues such as eye and mouth movements are effectively captured.
- Efficient Channel Attention: introduces a lightweight yet effective channel attention mechanism that achieves a favorable balance between classification accuracy and computational efficiency, making it suitable for real-time applications.
- Convolutional Block Attention Module: combines both channel and spatial attention in a unified manner, providing comprehensive feature refinement and enabling the network to capture complementary information across both dimensions.

By embedding these attention mechanisms into the CNN backbone, the framework allows for a systematic comparison of channel-only, spatial-only, lightweight, and hybrid attention strategies, thereby offering deeper insights into their relative effectiveness for real-time driver drowsiness detection.

To establish a fair reference point, a baseline CNN model without any attention mechanism is also implemented. This baseline serves as a control experiment, allowing us to quantify the direct contribution of each attention strategy when integrated into the architecture. By contrasting the performance of the baseline with its attention-augmented counterparts, the incremental value of channel, spatial, lightweight, and hybrid attention can be systematically evaluated. Furthermore, in order to position the proposed framework within the broader landscape of state-of-the-art approaches, we benchmark our models against six widely adopted transfer learning architectures: VGG19, InceptionV3, DenseNet169, MobileNet, ResNet50V2, and InceptionResNetV2. These architectures have consistently demonstrated high performance in computer vision tasks, including driver monitoring, and therefore provide a robust comparative baseline. By integrating both an internal benchmark (baseline CNN) and external benchmarks (transfer learning models), our study ensures a comprehensive evaluation of the effectiveness, efficiency, and practical applicability of attention mechanisms in driver drowsiness detection.

This systematic design enables not only the evaluation of different attention mechanisms but also the analysis of their trade-offs in terms of accuracy, computational complexity, and suitability for real-time deployment in ITS. As illustrated in Fig. 1, the proposed architecture follows a structured pipeline beginning with data acquisition and frame extraction, followed by facial landmark detection using DLib to isolate critical regions of interest (face, eyes, and mouth). These regions are preprocessed through resizing and normalization to ensure consistency across samples before being passed into the baseline CNN and its attention-augmented variants. In parallel, six state-of-the-art transfer learning models (VGG19, DenseNet169, ResNet50V2, InceptionV3, InceptionResNetV2, MobileNet) are fine-tuned to provide external benchmarks. Model outputs are evaluated using a comprehensive set of performance metrics, including accuracy, precision, recall, and F1-score, enabling a rigorous comparison across all candidate models. By integrating baseline, attention-enhanced, and transfer learning approaches into a unified evaluation framework, the proposed workflow provides a reliable and effective solution for real-time driver drowsiness detection.

A. OVERVIEW OF THE DATASET

This research utilizes the publicly available NTHU Driver Drowsiness Dataset (NTHU-DDD), developed by National Tsing Hua University, which has become one of the most widely adopted benchmarks for evaluating driver fatigue detection systems. The dataset contains extensive video recordings of drivers under varying conditions designed to simulate real-world drowsiness scenarios, including different head poses, facial occlusions, and lighting variations. An enhanced version of this dataset, referred to as NTHU-DDD2, has been released on Kaggle [29], where video frames have been pre-extracted and organized to facilitate direct use in deep learning workflows. While NTHU-DDD provides raw video data, NTHU-DDD2 builds upon it by supplying labeled image sequences that serve as the primary input for this study.

To ensure consistency and model robustness, all frames were subjected to standardized preprocessing steps. These included resizing to a fixed spatial resolution, normalization of pixel intensity values, and geometric transformations such as horizontal and vertical shifting, scaling, zooming, and rotation. Such preprocessing helps mitigate intra-class variability and prepares the data for efficient feature extraction. Furthermore, to reduce the risk of overfitting and to improve the generalizability of the models, data augmentation techniques were applied systematically across the dataset. This augmentation strategy not only balances class distributions but also exposes the models to a broader set of variations, enhancing their ability to recognize drowsiness cues under diverse real-world driving conditions.

The dataset leverages both facial and behavioral cues to facilitate reliable drowsiness detection. In particular, it contains diverse image samples that can be categorized into yawning, slow blinking with nodding, and combinations of these behaviors, all of which are well-established indicators of fatigue. Importantly, these categories are represented for drivers with and without glasses, thereby increasing the dataset's variability and robustness for real-world conditions. In total, the dataset consists of 66,521 grayscale images with a resolution of 640×480 pixels, divided into two classes: drowsy and not-drowsy. Specifically, 36,030 images belong to the drowsy category, while 30,491 images correspond to the not-drowsy category. Fig. 2 presents the dataset distribution across these two classes, illustrating a relatively balanced structure, which is crucial for preventing bias during model training. Representative examples of both drowsy and not-drowsy images are shown in Fig. 3, highlighting the visual diversity and behavioral patterns captured in the dataset.

The balanced representation of classes, combined with the inclusion of multiple fatigue indicators across varying driver conditions, makes this dataset highly suitable for evaluating and benchmarking deep learning-based drowsiness detection frameworks. Furthermore, its widespread use in prior studies reinforces its validity as a standard benchmark, enabling fair comparison with state-of-the-art approaches.

B. DLIB (FACE LANDMARKS DETECTION)

DLib is employed in this study as the primary tool for detecting facial landmarks, owing to its proven robustness and efficiency in real-time applications. DLib performs face detection by combining the Histogram of Oriented Gradients (HOG) descriptor with a linear SVM classifier [30]. The HOG descriptor captures the structural and edge information of the face, which is then classified by the SVM to determine the presence and location of facial regions. Once the face is detected, DLib applies a regression-based landmark localization model to extract predefined key points corresponding to essential facial components such as the eyes, mouth, and jawline.

Landmark detection refers to the identification of specific facial features from an image, which plays a critical role in subsequent tasks such as facial recognition, expression analysis, and behavior interpretation. In the context of driver drowsiness detection, precise localization of regions such as the eyes and mouth is particularly important, as micro-expressions like slow blinking, partial eyelid

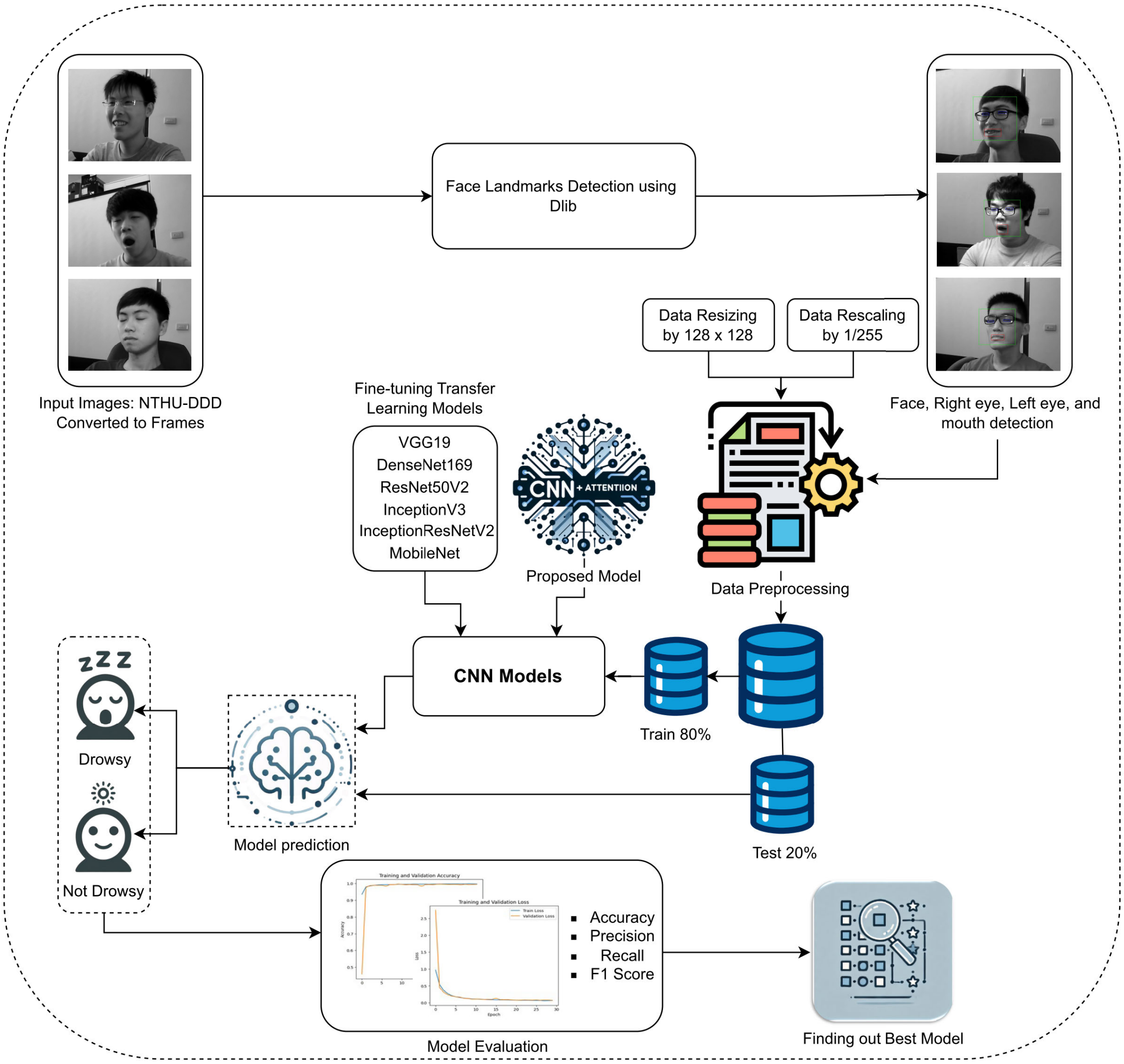


FIGURE 1. The proposed workflow architecture.

closure, and yawning are strong indicators of fatigue. Existing approaches to facial landmark detection typically fall into three categories: regression-based techniques, limited local model methods, and holistic strategies [31]. DLib adopts a regression-based approach, which efficiently maps image pixel intensities to landmark positions, striking a balance between accuracy and computational efficiency. This makes it well-suited for real-time vehicular applications where both reliability and speed are essential.

Holistic approaches model the face as a whole entity and fit a global template to an input image. A well-known example is the Active Appearance Model (AAM), which jointly represents both facial shape and appearance variations. In AAM, the objective is to optimize a set of model parameters by minimizing the difference between the fitted model and the target facial image. While such methods are computationally efficient and demonstrate strong performance under controlled laboratory conditions, they suffer from critical limitations in real-world scenarios. In particular, variations in lighting, facial expressions, and head poses significantly degrade their accuracy, making them less reliable for tasks such as driver monitoring. Moreover, holistic methods are prone to misalignment when portions of the face are occluded, such as when drivers wear glasses or cover their mouth. These limitations underscore the need for more robust and adaptive approaches, such as regression-based facial landmark detection in DLib, which is better suited for handling unconstrained environments and ensuring reliable localization of key facial regions essential for drowsiness detection.

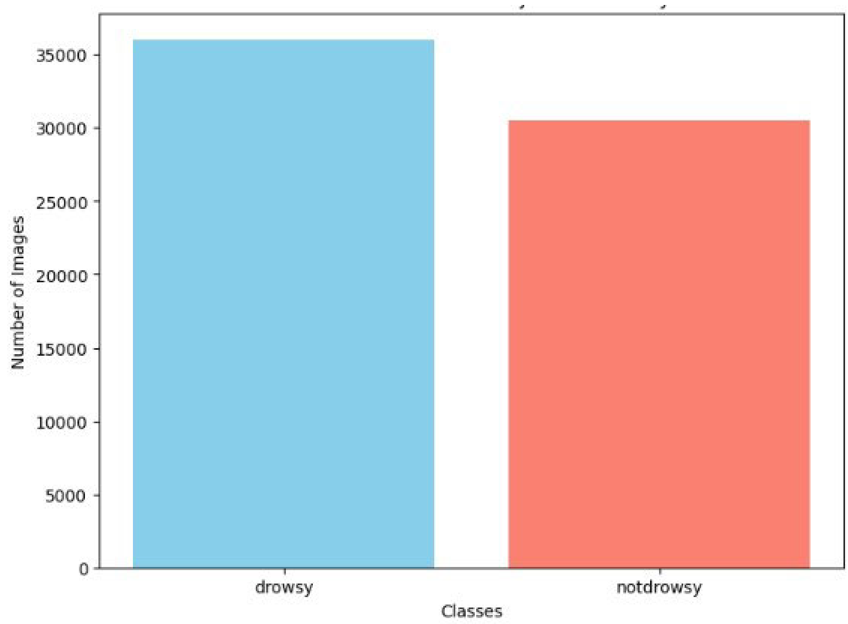


FIGURE 2. Dataset distribution: Drowsy vs Notdrowsy.

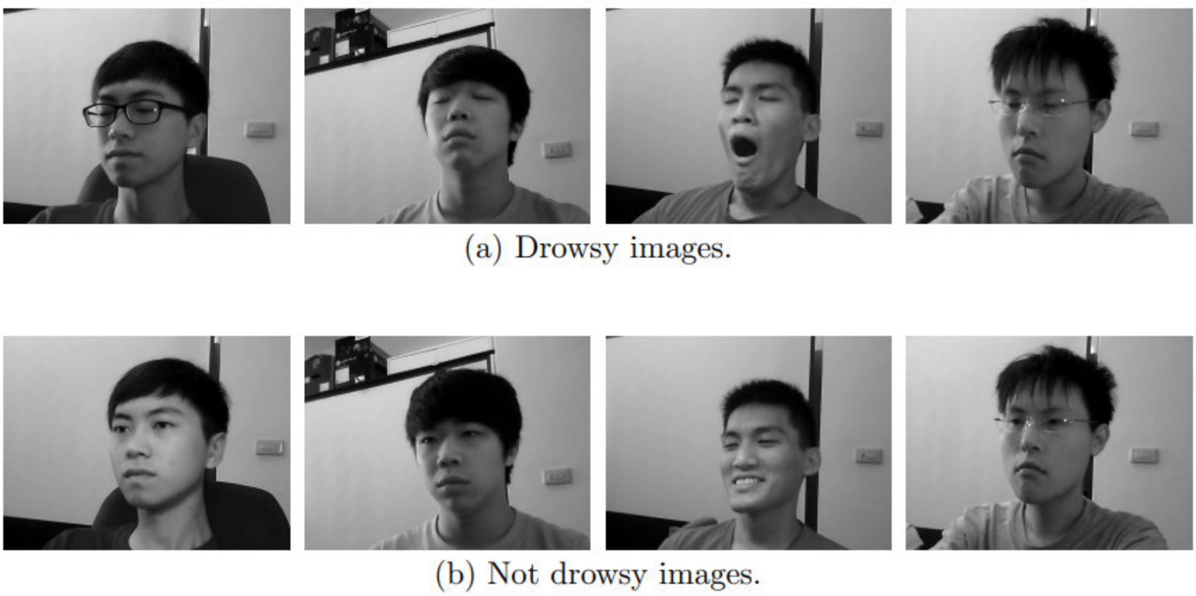
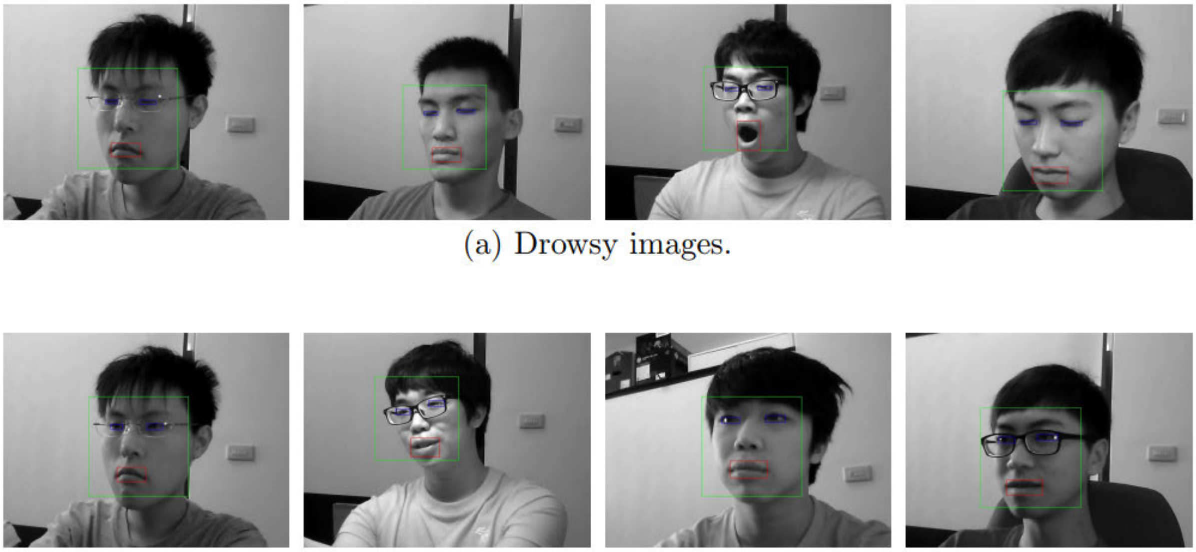


FIGURE 3. Sample images: (a) Drowsy images, (b) Not drowsy images.

Constrained Local Model (CLM) methods focus on localized regions of the face rather than attempting to fit a global template to the entire facial structure. One of the most widely adopted techniques in this category is the Active Shape Model (ASM), which aligns a statistical model of face shape with individual facial features by leveraging the neighborhood appearance of each landmark. These methods incorporate shape deformation constraints to guide the search for the optimal landmark positions, thereby achieving improved robustness to minor variations in facial expression and pose compared to holistic approaches. However, CLM-based methods still encounter significant challenges when faced with extensive pose variations, strong illumination changes, or partial occlusions. While they represent a step forward in comparison to holistic models, their performance in highly dynamic and unconstrained environments, such as real-world driving scenarios, remains limited. These shortcomings further justify the adoption of regression-based



(a) Drowsy images.

(b) Not drowsy images.

FIGURE 4. Sample images after using Dlib to detect face landmarks (a) Drowsy images, (b) Not drowsy images.

landmark detection strategies, as implemented in DLib, which provide greater flexibility and reliability for capturing critical facial cues essential for accurate driver drowsiness detection.

Regression-based methods propose learning a direct mapping from raw image pixels to landmark coordinates through data-driven machine learning techniques. Unlike holistic or local models, which depend on predefined geometrical assumptions, regression-based approaches bypass explicit modeling and instead predict landmark positions directly. Modern implementations typically leverage deep CNNs, which excel at learning hierarchical feature representations and are capable of localizing facial landmarks with high accuracy, even under challenging conditions such as illumination variation, occlusion, or pose changes.

In this study, we employ DLib's face detector and shape predictor, which utilize a regression-based pipeline for facial landmark detection [32]. The model predicts landmark positions by iteratively refining estimates through cascaded regression, ensuring both robustness and computational efficiency. Using this approach, bounding boxes are generated around the most critical regions of interest, the left eye, right eye, mouth, and overall face, as shown in Fig. 4. These localized regions provide the foundation for subsequent feature extraction, enabling the proposed framework to focus on the subtle visual cues that distinguish drowsy states from alert ones.

C. PREPARATION AND PROCESSING OF DATA

Data preparation and preprocessing represent essential steps in deep learning workflows, as the quality and consistency of input data directly influence the stability and performance of the model. Effective preprocessing reduces the risk of overfitting, accelerates the training process, and ensures that the model generalizes effectively to unseen data. In the context of computer vision, preprocessing is particularly critical because raw image data often varies in resolution, illumination, and pixel intensity distribution. Standardizing these inputs allows models to focus on meaningful feature representations rather than irrelevant variations.

For this study, all images were resized to a fixed spatial resolution of 128×128 pixels. This resizing ensures uniformity across the dataset while also reducing computational overhead, thereby facilitating faster model training and inference. In addition, pixel intensity normalization was applied by dividing each pixel value by 255, effectively rescaling values into the range [0, 1]. Normalization reduces the impact of raw pixel magnitude differences, leading to improved numerical stability during gradient updates and enabling the model to converge more efficiently. Together, resizing and normalization form a crucial foundation for subsequent feature extraction, ensuring that the proposed framework can effectively capture discriminative patterns associated with drowsy and non-drowsy states.

D. ATTENTION MECHANISMS

Attention mechanisms have emerged as powerful components in modern deep learning architectures, enabling models to selectively emphasize the most informative features while suppressing irrelevant or redundant information. Unlike conventional convolutional operations that treat all features equally, attention modules provide adaptive weighting, allowing the network to focus on subtle yet discriminative cues. This is particularly valuable in driver drowsiness detection, where critical indicators such as partial eyelid closure, prolonged blinking, or yawning may otherwise be overshadowed by background noise or non-informative facial details. By guiding the model's focus toward the most relevant spatial regions or feature channels, attention mechanisms enhance the robustness and interpretability of the learning process.

In this work, we integrate and systematically evaluate four well-established attention strategies within our CNN-based framework:

- Squeeze-and-Excitation: a channel attention mechanism that adaptively recalibrates feature maps by modeling interdependencies between channels, thereby improving representational power.
- Spatial Attention: focuses on emphasizing critical spatial locations in feature maps, ensuring that localized drowsiness-related patterns such as eye or mouth movements are effectively captured.
- Efficient Channel Attention: introduces a lightweight design that achieves competitive performance with minimal computational overhead, striking a balance between accuracy and efficiency.
- Convolutional Block Attention Module: a hybrid approach that sequentially applies both channel and spatial attention, offering a more comprehensive refinement of features for improved detection performance.

Together, these mechanisms allow for a systematic comparative analysis across channel-based, spatial-based, lightweight, and hybrid strategies, offering deeper insights into their respective contributions for real-time driver drowsiness detection.

1) SQUEEZE-AND-EXCITATION

CNNs employ the SE block to enhance the feature extraction process and increase the representational capacity of the model [33], [34]. By explicitly modeling inter-channel dependencies, the SE block adaptively recalibrates feature maps to emphasize the most informative channels while suppressing less relevant ones. This mechanism is lightweight, modular, and can be seamlessly integrated into a wide range of CNN architectures, consistently yielding notable performance improvements across computer vision tasks [35], [36], [37]. In the context of driver drowsiness detection, the SE block is particularly valuable for highlighting fine-grained visual cues such as subtle eye closures or mouth movements, which may otherwise be overlooked in standard convolutional feature extraction.

a: INPUT FEATURE MAPS

The SE block operates on the output feature maps generated by a convolutional layer. Let the feature map be represented as $X \in \mathbb{R}^{H \times W \times C}$, where H, W, and C denote the spatial height, width, and number of channels, respectively. Each channel in X can be interpreted as a specific feature detector, but conventional CNNs treat all channels equally, without considering their relative importance. The SE block addresses this limitation by introducing a mechanism to adaptively rescale channels based on their global contribution to the task at hand.

b: SQUEEZE OPERATION

The squeeze step aims to capture global spatial information by reducing each feature map into a single scalar value. This is achieved using Global Average Pooling (GAP), which computes the mean activation of each channel across its spatial dimensions. By doing so, the network condenses the two-dimensional spatial features into a one-dimensional channel descriptor, thereby retaining essential global context while discarding redundant spatial information. Mathematically, for each channel c, the squeeze operation is defined as:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{i,j,c}, \tag{1}$$

where $x_{i,j,c}$ denotes the activation at position (i, j) in channel c, while H and W represent the height and width of the feature map, respectively. The output of this operation is a vector $\mathbf{z} \in \mathbb{R}^C$, where C is the number of channels. This vector serves as a compact representation of channel-level importance.

c: EXCITATION OPERATION

The excitation step models the interdependencies between channels and learns adaptive weights that indicate their relative importance. The vector $\mathbf{z}$ from the squeeze operation is passed through a bottleneck architecture composed of two fully connected (FC) layers. The first FC layer reduces dimensionality to C/r, where r is a reduction ratio that controls model complexity, followed by a ReLU activation. The second FC layer restores the dimensionality to C, after which a sigmoid activation function normalizes the output to the range $[0, 1]$. The excitation operation is formally expressed as:

$$\mathbf{s} = \sigma\left(W_2 \cdot \mathrm{ReLU}(W_1 \cdot \mathbf{z})\right), \tag{2}$$

where $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ are the learnable weight matrices of the two FC layers, ReLU denotes the rectified linear activation, and $\sigma(\cdot)$ represents the sigmoid function. The output $\mathbf{s} \in \mathbb{R}^C$ is a vector of channel-wise weights, with each element indicating the significance of its corresponding feature channel.

d: RECALIBRATION

In the final stage, the channel-wise weights $\mathbf{s}$ obtained from the excitation step are used to recalibrate the original feature maps. This is achieved through element-wise multiplication, where each channel in the feature map is rescaled according to its learned importance. Channels assigned higher weights are amplified, while those with lower weights are suppressed, thereby refining the overall feature representation. Formally, the recalibrated output $\hat{F} \in \mathbb{R}^{H \times W \times C}$ is computed as:

$$\hat{F}_c = s_c \cdot F_c, \quad c = 1, 2, \ldots, C, \tag{3}$$

where F_c denotes the c^{th} channel of the input feature map, s_c is the corresponding weight from the excitation step, and $\hat{F}_c$ represents the recalibrated channel.

This operation ensures that the network adaptively emphasizes the most informative channels while diminishing the contribution of less relevant ones. In the context of driver drowsiness detection, recalibration allows the model to prioritize subtle but critical cues, such as eyelid closures or yawning, while reducing the influence of background noise or irrelevant features.

2) SPATIAL ATTENTION (SA)

Spatial attention mechanisms aim to highlight the most informative regions within a feature map by directing the network's focus to the spatial locations that contribute most significantly to the learning task [38]. In contrast to channel attention, which determines the relative importance of feature channels ("what" to focus on), spatial attention emphasizes the localization aspect of feature representation ("where" to focus). This distinction makes SA particularly relevant for tasks such as driver drowsiness detection, where fine-grained spatial patterns—such as eyelid closure, yawning, or subtle head tilts, must be localized precisely within the driver's face.

a: INPUT FEATURE PROCESSING

Given an input feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, spatial attention generates an attention map $\mathbf{M}_s \in \mathbb{R}^{H \times W \times 1}$ that captures the importance of each spatial position across all channels. To aggregate spatial context information, pooling operations are typically applied along the channel dimension. Specifically, both average pooling and max pooling are employed to compute two separate spatial context descriptors, which are then concatenated to form a representative feature:

$$\mathbf{X}_s = \left[\text{AvgPool}(\mathbf{X});\ \text{MaxPool}(\mathbf{X})\right], \tag{4}$$

where $\mathbf{X}_s \in \mathbb{R}^{H \times W \times 2}$ is the concatenated descriptor. This representation preserves essential information regarding the distribution of activations across spatial positions, enabling the network to differentiate between informative regions (e.g., eyes or mouth) and less relevant background areas.

b: CHANNEL INFORMATION AGGREGATION

In the first stage of spatial attention, information across all channels is aggregated to generate spatial descriptors that highlight the importance of each location in the feature map. This is accomplished through two complementary pooling operations applied along the channel dimension: average pooling and max pooling. Average pooling captures the overall distribution of activations, reflecting the general importance of spatial positions, whereas max pooling emphasizes the most dominant responses, thereby preserving salient local features. Formally, given an input feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, the pooled spatial descriptors are defined as:

$$\mathbf{F}_{avg} = \frac{1}{C} \sum_{c=1}^{C} \mathbf{X}_{:,:,c}, \tag{5}$$

$$\mathbf{F}_{max} = \max_{c=1}^{C} \mathbf{X}_{:,:,c}, \tag{6}$$

where $\mathbf{F}_{avg}, \mathbf{F}_{max} \in \mathbb{R}^{H \times W}$ represent the average-pooled and max-pooled feature maps, respectively. These two descriptors complement each other, with $\mathbf{F}_{avg}$ providing a smoother global representation of feature activations and $\mathbf{F}_{max}$ retaining sharp, high-contrast responses. Together, they serve as the foundation for constructing the spatial attention map.

c: SPATIAL ATTENTION MAP GENERATION

Once the average-pooled and max-pooled descriptors are obtained, they are concatenated along the channel dimension to form a combined spatial feature representation. This joint descriptor is then passed through a convolutional layer with a kernel size of 7 × 7, which is sufficiently large to capture broader contextual relationships across neighboring pixels while maintaining computational efficiency. The output of this convolution is normalized by a sigmoid activation function to constrain the attention values to the range [0, 1], thereby producing a spatial attention map $\mathbf{M}_s \in \mathbb{R}^{H \times W \times 1}$. Formally, the process can be expressed as:

$$\mathbf{M}_s = \sigma\left(\text{Conv}^{7\times7}\left([\mathbf{F}_{avg}; \mathbf{F}_{max}]\right)\right), \tag{7}$$

where $[\mathbf{F}_{avg}; \mathbf{F}_{max}]$ denotes concatenation along the channel dimension, $\text{Conv}^{7\times7}$ is the convolutional operation with a kernel size of 7×7, and $\sigma(\cdot)$ represents the sigmoid activation. The resulting attention map assigns an importance weight to each spatial location, highlighting informative regions such as the eyes and mouth while down-weighting less relevant background areas.

d: SPATIAL RECALIBRATION

In the final step, the spatial attention map $\mathbf{M}_s$ is applied to the original input feature map $\mathbf{X}$ through element-wise multiplication. This operation adaptively reweights each spatial location according to its learned importance, amplifying discriminative regions (e.g., eyes and mouth) while suppressing less informative background areas. The recalibrated feature map $\tilde{\mathbf{X}} \in \mathbb{R}^{H \times W \times C}$ is formally expressed as:

$$\tilde{\mathbf{X}} = \mathbf{M}_s \odot \mathbf{X}, \tag{8}$$

where $\odot$ denotes element-wise multiplication applied across all spatial positions and extended uniformly across the C channels. This recalibration ensures that subsequent convolutional layers receive spatially refined features, enabling the network to focus more effectively on subtle driver drowsiness indicators such as eyelid closure, blinking frequency, and yawning.

3) EFFICIENT CHANNEL ATTENTION

The ECA module [39] was proposed as an improvement over the SE block, with the primary goal of reducing computational complexity while maintaining or even enhancing representational effectiveness. Unlike SE, which employs dimensionality reduction through fully connected layers to model channel dependencies, ECA eliminates this step and instead captures local cross-channel interactions using a lightweight one-dimensional convolution. This design not only preserves channel-wise information more effectively but also avoids the performance degradation that can result from dimensionality reduction.

a: INPUT FEATURE ANALYSIS

Given an input feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, global average pooling is first applied to aggregate spatial information into a compact channel descriptor:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{i,j,c}, \quad c = 1, 2, \ldots, C, \tag{9}$$

where $x_{i,j,c}$ denotes the activation at spatial position (i, j) of channel c. This results in a descriptor vector $\mathbf{z} \in \mathbb{R}^C$, which summarizes the global contextual information of each channel. Instead of passing this descriptor through fully connected layers, ECA applies a one-dimensional convolution to capture local dependencies between neighboring channels.

b: GLOBAL AVERAGE POOLING

The first step of ECA involves computing channel-wise statistics using GAP, which condenses spatial information into a compact channel descriptor without dimensionality reduction. For each channel c, the pooled descriptor is computed as:

$$\mathbf{z}_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c(i, j), \tag{10}$$

where $x_c(i, j)$ represents the activation at spatial location (i, j) of channel c. The resulting vector $\mathbf{z} \in \mathbb{R}^C$ summarizes the global contextual information of all C channels.

c: LOCAL CROSS-CHANNEL INTERACTION

Unlike SE blocks, which rely on fully connected layers to model channel dependencies, ECA employs a more efficient strategy by applying a one-dimensional convolution directly to the descriptor $\mathbf{z}$. This operation captures local cross-channel dependencies by allowing each channel to interact only with its k nearest neighbors, thereby reducing complexity while preserving representational power. The attention weights are obtained as:

$$\mathbf{w} = \sigma\left(\mathrm{Conv1D}^k(\mathbf{z})\right), \tag{11}$$

where $\mathbf{w} \in \mathbb{R}^C$ represents the channel attention weights, $\mathrm{Conv1D}^k$ denotes a one-dimensional convolution with kernel size k, and $\sigma(\cdot)$ is the sigmoid activation function.

The kernel size k is not fixed but adaptively determined as a function of the channel dimension C, ensuring that larger networks capture wider cross-channel dependencies while smaller ones focus on narrower interactions:

$$k = \psi(C) = \left| \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right|_{\text{odd}}, \tag{12}$$

where $\gamma = 2$ and $b = 1$ are empirically chosen parameters, and $| \cdot |_{\text{odd}}$ denotes rounding to the nearest odd integer. This adaptive mechanism allows ECA to scale efficiently with varying network widths, maintaining a balance between computational efficiency and representational effectiveness.

d: EFFICIENT RECALIBRATION

In the final stage, the channel attention weights $\mathbf{w}$ obtained from the local cross-channel interaction step are applied to the original feature map $\mathbf{X}$. Unlike SE blocks, which require dimensionality reduction and restoration, ECA performs direct element-wise multiplication, thereby avoiding additional computational overhead. Each channel is rescaled according to its learned weight, while the spatial resolution of the feature map remains unchanged. The recalibrated feature map $\tilde{\mathbf{X}} \in \mathbb{R}^{H \times W \times C}$ is computed as:

$$\tilde{\mathbf{X}} = \mathbf{w} \odot \mathbf{X}, \tag{13}$$

where $\odot$ denotes element-wise multiplication, such that each weight w_c is broadcast across all spatial positions of the corresponding channel X_c. This operation ensures that informative channels, such as those capturing eye closure or yawning cues, are emphasized, while less relevant channels are suppressed, ultimately improving the model's robustness for real-time driver drowsiness detection.

4) CONVOLUTIONAL BLOCK ATTENTION MODULE

The CBAM [40] represents a hybrid attention mechanism that integrates both channel and spatial attention in a sequential manner. Its design leverages the complementary strengths of these two attention types: channel attention emphasizes "what" features are important, while spatial attention highlights "where" these features are most relevant. By combining both dimensions, CBAM provides a more comprehensive feature refinement strategy, enabling the network to capture subtle yet critical patterns in visual data.

a: HYBRID ATTENTION FRAMEWORK

Given an input feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$, CBAM applies attention in two successive stages. First, channel attention $\mathbf{M}_c \in \mathbb{R}^{1 \times 1 \times C}$ is computed and used to refine the input feature map along the channel dimension. Next, spatial attention $\mathbf{M}_s \in \mathbb{R}^{H \times W \times 1}$ is applied to the channel-refined features, further enhancing the most informative spatial locations. The overall CBAM output can be expressed as:

$$\tilde{\mathbf{F}} = \mathbf{M}_s \odot (\mathbf{M}_c \odot \mathbf{F}), \tag{14}$$

where $\odot$ denotes element-wise multiplication with appropriate broadcasting. This sequential design ensures that the model first identifies the most critical feature channels and subsequently focuses on the most relevant spatial regions, leading to a more discriminative and robust representation.

b: CHANNEL ATTENTION STAGE

The first stage of CBAM focuses on identifying the most informative feature channels by applying channel attention. To achieve this, two complementary channel descriptors are extracted from the input feature map $\mathbf{F}$ using both global average pooling and global max pooling. Average pooling captures the overall contextual distribution of activations, while max pooling preserves the most salient responses, ensuring that both global trends and critical local features are considered.

The pooled descriptors are then passed through a shared multilayer perceptron (MLP), consisting of two fully connected layers with a hidden dimension of $\frac{C}{r}$, where r is the reduction ratio used to control complexity. The outputs from the average- and max-pooled paths are summed and normalized using a sigmoid activation function to produce the channel attention map $\mathbf{M}_c(\mathbf{F}) \in \mathbb{R}^{1\times 1\times C}$. Formally,

$$\mathbf{M}_c(\mathbf{F}) = \sigma\left(\mathrm{MLP}(\mathrm{AvgPool}(\mathbf{F})) + \mathrm{MLP}(\mathrm{MaxPool}(\mathbf{F}))\right), \tag{15}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function. This attention map is then applied to the original input feature map via element-wise multiplication:

$$\mathbf{F}' = \mathbf{M}_c(\mathbf{F}) \odot \mathbf{F}. \tag{16}$$

Through this recalibration, CBAM selectively emphasizes feature channels that capture critical visual cues (such as eyelid closures, yawning, or head tilts), while suppressing less informative channels, thereby improving the model's ability to detect subtle drowsiness indicators.

c: SPATIAL ATTENTION STAGE

Following the refinement of channel-wise features, CBAM applies spatial attention to emphasize the most informative regions within the feature map. Unlike channel attention, which determines "what" features are important, spatial attention identifies "where" these features are located in the spatial domain.

To construct the spatial attention map, pooling operations are performed along the channel dimension of the channel-refined feature map $\mathbf{F}'$. Specifically, both average pooling and max pooling are applied to aggregate spatial descriptors, with average pooling capturing smooth contextual information and max pooling preserving high-contrast salient details. The two resulting feature maps are concatenated and passed through a convolutional layer with a 7×7 kernel to capture local contextual dependencies across a sufficiently wide receptive field. Finally, a sigmoid activation function normalizes the output to the range [0, 1], producing the spatial attention map $\mathbf{M}_s(\mathbf{F}') \in \mathbb{R}^{H\times W\times 1}$. Formally:

$$\mathbf{M}_s(\mathbf{F}') = \sigma\left(\mathrm{Conv}^{7\times 7}\left([\mathrm{AvgPool}(\mathbf{F}');\ \mathrm{MaxPool}(\mathbf{F}')]\right)\right), \tag{17}$$

The resulting spatial attention map is then applied to the channel-refined features $\mathbf{F}'$ via element-wise multiplication:

$$\mathbf{F}'' = \mathbf{M}_s(\mathbf{F}') \odot \mathbf{F}', \tag{18}$$

where $\odot$ denotes element-wise multiplication with broadcasting. This recalibration ensures that the network prioritizes spatial regions that are highly indicative of drowsiness, such as partially closed eyes or yawning mouths, while suppressing irrelevant background areas. By sequentially combining channel and spatial attention, CBAM yields a more discriminative and robust feature representation for real-time driver monitoring.

d: COMBINED ATTENTION OUTPUT

The complete CBAM operation is realized by sequentially applying channel attention followed by spatial attention. This design ensures that the network first identifies "what" feature channels are most important and subsequently determines "where" these critical features are located within the spatial domain. Formally, the combined operation can be expressed as:

$$\tilde{\mathbf{X}} = \mathbf{M}_s\left(\mathbf{M}_c(\mathbf{X}) \odot \mathbf{X}\right) \odot \left(\mathbf{M}_c(\mathbf{X}) \odot \mathbf{X}\right), \tag{19}$$

where $\mathbf{M}_c(\mathbf{X}) \in \mathbb{R}^{1\times 1\times C}$ and $\mathbf{M}_s(\cdot) \in \mathbb{R}^{H\times W\times 1}$ denote the channel and spatial attention maps, respectively. The output $\tilde{\mathbf{X}}$ represents a feature map that has been refined along both the channel and spatial dimensions.

By combining the complementary benefits of channel and spatial recalibration, CBAM produces highly discriminative representations capable of capturing subtle drowsiness cues, such as gradual eyelid closure, yawning, and micro head movements. This sequential attention design makes CBAM particularly effective for complex visual recognition tasks, including real-time driver monitoring in challenging environments.

E. CNN MODELS ARCHITECTURES

CNNs serve as the foundation of our experimental framework for driver drowsiness detection. Their ability to capture hierarchical spatial features makes them particularly well-suited for identifying subtle facial and behavioral cues such as blinking, yawning, and nodding. In this section, we describe the architectural designs employed in our study, beginning with a baseline CNN model that serves as a reference point for performance evaluation. We then introduce our proposed attention-augmented CNN architecture, which integrates advanced attention mechanisms to enhance feature discrimination and robustness. Together, these models provide a systematic basis for assessing the role of attention in improving the accuracy and efficiency of drowsiness detection systems.

1) PROPOSED MODEL

To enhance the efficiency and reliability of image classification for driver fatigue detection, we propose a novel CNN architecture augmented with attention mechanisms. The central motivation is that traditional CNNs, although effective in extracting spatial hierarchies, may fail to emphasize subtle yet critical cues of drowsiness such as gradual eyelid closure, yawning, or slight head tilts. By integrating attention modules, the model adaptively prioritizes the most informative features while suppressing irrelevant background information, thereby improving discriminative power.

Specifically, the proposed architecture incorporates four variants of attention mechanisms: SE, SA, ECA, and the hybrid CBAM. Each of these modules is strategically embedded into the CNN pipeline to strengthen feature representation at different stages of learning.

The model consists of two primary components: convolutional blocks and attention blocks. The convolutional blocks are responsible for hierarchical feature extraction through stacked convolutional layers, each followed by non-linear activation functions and pooling operations for spatial dimensionality reduction. The attention blocks are interleaved with these convolutional layers, ensuring that both channel-level and spatial-level recalibration occur during feature learning.

The input to the network is standardized grayscale images resized to 128×128 pixels. After passing through successive convolutional and attention blocks, the feature maps are flattened and fed into fully connected layers that perform the final binary classification: "drowsy" versus "not drowsy." By combining the strengths of convolutional operations with diverse attention strategies, the proposed CNN is optimized for both accuracy and computational efficiency, making it suitable for real-time deployment in intelligent transportation systems.

The attention modules incorporated in our framework each perform distinct yet complementary roles in refining feature representations. The SE module captures global context by reducing each feature map to a single descriptor via global average pooling, followed by two fully connected layers that model inter-channel dependencies. The excitation operation applies a bottleneck transformation with ReLU activation for dimensionality reduction, and then restores the channel dimension with a sigmoid activation to generate channel-specific weights. These weights are subsequently applied in the recalibration step, allowing the network to selectively emphasize informative channels while suppressing less relevant ones.

In contrast, the SA mechanism focuses on spatial dependencies by aggregating channel information through average and max pooling, followed by a convolutional layer that produces a spatial attention map. This enables the model to highlight critical regions within feature maps, such as eyes or mouth regions, that are strongly indicative of drowsiness. The ECA module offers a lightweight alternative to SE by eliminating dimensionality reduction and instead employing local cross-channel interactions via a one-dimensional convolution. This design maintains computational efficiency while still effectively capturing channel relationships. Finally, the CBAM integrates both channel and spatial attention sequentially, thereby combining the strengths of SE and SA in a unified architecture.

Together, these attention mechanisms enhance the discriminative capacity of CNNs by directing focus toward the most salient features, whether along the channel or spatial dimensions, while maintaining computational efficiency. This synergy is particularly valuable for driver drowsiness detection, where subtle visual cues must be reliably identified in real time.

Each convolutional block in the proposed architecture consists of a convolutional layer with a 3×3 kernel, designed with appropriate padding to preserve spatial dimensions, followed by batch normalization to stabilize and accelerate convergence, and a ReLU activation function to introduce non-linearity and improve learning efficiency. Max pooling with a 2×2 window is employed to progressively reduce the spatial resolution while retaining the most salient features. To further enhance the representational capacity of these features, an attention block is inserted after each convolutional block. These attention modules adaptively emphasize the most informative feature channels or spatial regions, thereby improving the network's ability to capture subtle cues of drowsiness such as partial eye closure, yawning, or head tilts.

The architecture, illustrated in Fig. 5, is organized into three main stages, each consisting of a convolutional block paired with an attention block. After these stages, the feature maps are flattened and passed through a fully connected layer with 512 units, followed by a dropout layer to mitigate overfitting. The final output layer employs a sigmoid activation function to classify the input as "drowsy" or "not drowsy." The network is trained using a suitable optimizer and binary cross-entropy loss function, with performance evaluated using accuracy, precision, recall, and F1-score metrics. This systematic design ensures not only high classification accuracy but also computational efficiency, making the architecture well-suited for deployment in real-time intelligent transportation systems.

2) FINE-TUNING TRANSFER LEARNING MODELS

Transfer learning has emerged as a powerful paradigm in computer vision, enabling the adaptation of large-scale pretrained models to specialized tasks with limited domain-specific data. Popular deep architectures such as VGG19, MobileNet, DenseNet169, InceptionV3, ResNet50V2, and InceptionResNetV2, originally trained on the extensive ImageNet dataset, have demonstrated exceptional performance across tasks including image classification, object detection, and feature extraction. Their success lies in their ability to learn hierarchical and transferable feature representations that generalize well to new domains.

In this study, these six architectures are fine-tuned and benchmarked against our proposed CNN with attention mechanisms for the task of driver drowsiness detection.

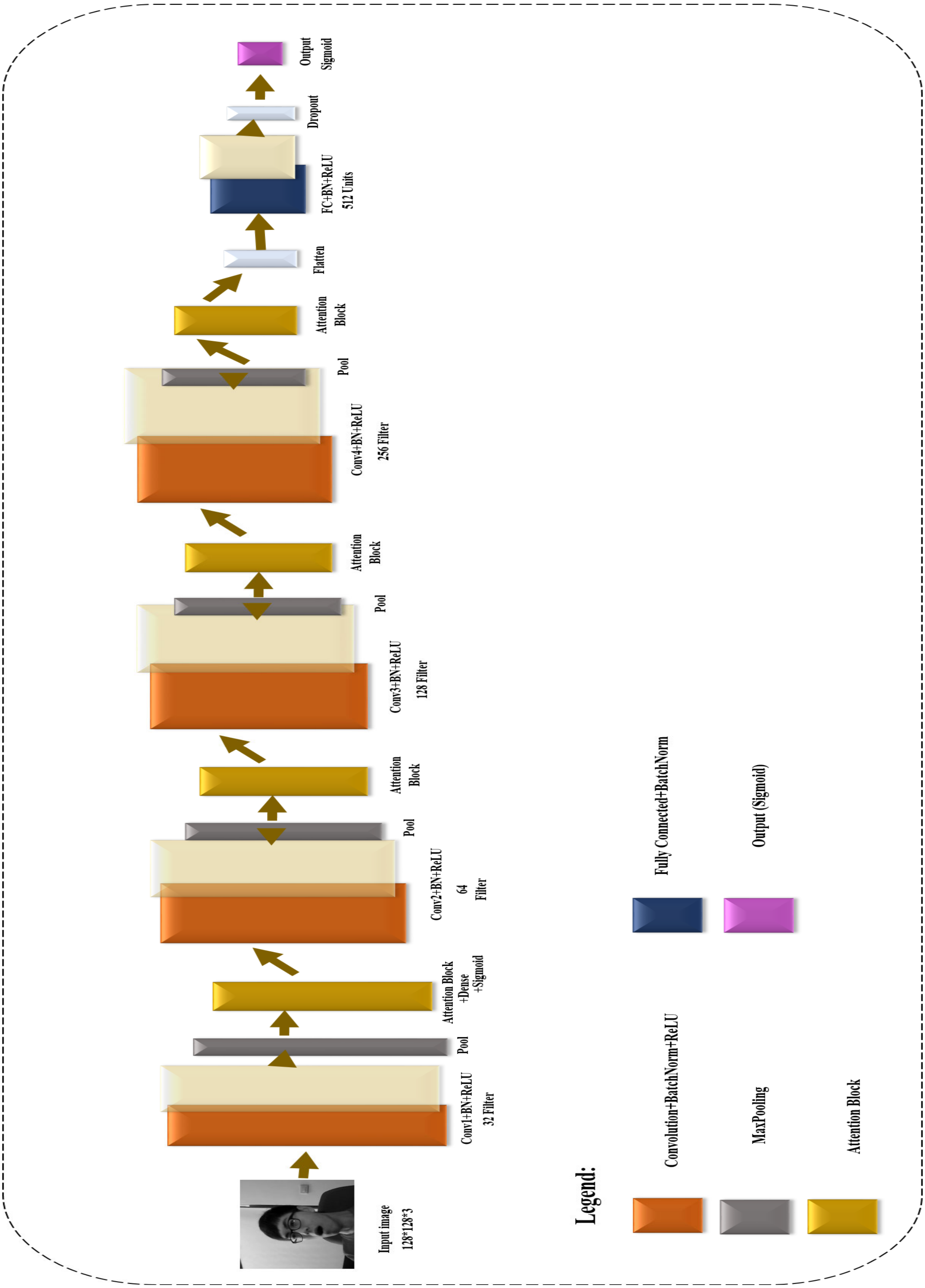


FIGURE 5. Proposed CNN architecture enhanced with SE, Spatial, ECA, and CBAM attention modules.

The models were selected to represent a broad spectrum of design philosophies: VGG19 as a deep yet relatively simple convolutional network; MobileNet as a lightweight and computationally efficient model suitable for real-time deployment; DenseNet169 for its densely connected layers that promote feature reuse; InceptionV3 for its multi-scale feature extraction; ResNet50V2 for its residual connections that address vanishing gradients in deep networks; and InceptionResNetV2, which combines Inception modules with residual learning for enhanced accuracy. A detailed overview of each model is provided in the following subsections.

VGG19 is a Deep Convolutional Neural Network (DCNN) widely recognized for its simplicity and effectiveness in large-scale image classification tasks [41], [42]. The network comprises sixteen convolutional layers followed by three fully connected layers, giving it a total of nineteen trainable layers from which its name is derived. Its design philosophy emphasizes the repeated use of small 3×3 convolutional filters and max-pooling layers, which enable the capture of fine-grained spatial details while progressively reducing feature map dimensions. The fully connected layers serve to integrate the extracted features into a global representation, while the final softmax output layer performs the classification across the target categories.

One of VGG19's major strengths lies in its uniform architecture, which makes it computationally straightforward to implement and highly effective at hierarchical feature extraction. Despite these advantages, the network's large number of parameters results in high memory consumption and longer inference times, which can be limiting for real-time applications. Nevertheless, VGG19 remains a popular benchmark model in transfer learning, and in the context of driver drowsiness detection, its ability to capture subtle spatial features such as eye closure or yawning movements makes it a valuable comparator against more modern and lightweight architectures.

DenseNet169 is a deep convolutional architecture belonging to the DenseNet family, consisting of 169 layers in total. Unlike traditional feedforward CNNs, DenseNets introduce dense connectivity patterns, where each layer receives inputs from all preceding layers and passes its output to all subsequent layers [43]. These direct connections, known as dense links, ensure maximum feature reuse across the network, thereby improving parameter efficiency and strengthening gradient propagation.

The architectural design of DenseNet addresses two major challenges in deep learning: the risk of overfitting due to redundant parameters and the vanishing gradient problem in very deep networks [44], [45]. By reusing previously computed feature maps, DenseNet significantly reduces the number of parameters compared to similarly deep architectures, while also facilitating more robust feature learning. This leads to improved accuracy and training efficiency without sacrificing representational power.

In the context of driver drowsiness detection, DenseNet169's ability to capture fine-grained variations in facial dynamics while maintaining computational efficiency makes it particularly well-suited. The reuse of features allows the model to detect subtle cues such as micro-expressions, partial eye closures, or yawning patterns more effectively, offering advantages over traditional CNNs that may overlook these nuanced details.

ResNet50V2 is an extension of the ResNet50 model, a member of the Residual Networks (ResNet) family, designed to enable the training of very deep neural networks [46]. ResNet50 itself consists of 50 layers and introduces the concept of residual learning through skip connections, where outputs from earlier layers are directly added to later layers. These identity shortcuts allow gradients to flow more effectively during backpropagation, thereby mitigating the vanishing gradient problem that typically hampers deep networks [47].

ResNet50V2 improves upon the original ResNet50 by modifying the internal block design. Specifically, it introduces pre-activation residual blocks, where batch normalization and ReLU activation are applied before the convolutional operations rather than after. This adjustment stabilizes training, enhances optimization, and leads to better generalization in many scenarios. Furthermore, the network employs both convolutional and identity blocks, offering a balance between computational efficiency and depth.

The robustness of ResNet architectures, coupled with their ability to preserve information across layers, has made them highly effective for various computer vision tasks. In the context of driver drowsiness detection, ResNet50V2's residual connections ensure that subtle spatial features, such as partial eye closure, blinking frequency, or micro head tilts, are captured consistently across multiple layers. Its stability and representational power make it a strong candidate for benchmarking against specialized architectures that incorporate attention mechanisms.

InceptionV3 is the third major version of Google's Inception Network, also known as GoogLeNet, and represents a highly optimized convolutional neural architecture for image classification [48]. The central innovation of Inception networks lies in the use of *Inception modules*, which consist of multiple parallel convolutional and pooling operations, typically 1×1, 3×3, and 5×5 convolutions, combined with max or average pooling. The outputs from these parallel operations are concatenated, enabling the network to simultaneously capture both fine-grained and coarse-grained features at multiple scales.

InceptionV3 extends this design with several optimizations, including the use of factorized convolutions (e.g., replacing a 5×5 convolution with two successive 3×3 convolutions) to reduce computational complexity, as well as auxiliary classifiers that act as regularizers and improve gradient flow during training. With a total depth of 42 layers, InceptionV3 achieves a strong balance between accuracy and efficiency, making it suitable for large-scale image recognition tasks while maintaining manageable computational requirements.

For driver drowsiness detection, the ability of InceptionV3 to capture features at multiple spatial resolutions is particularly advantageous. Subtle facial cues such as micro-expressions, partial eyelid closures, or mouth movements can be effectively detected across different receptive field sizes, thereby enhancing the robustness of the model under varying lighting conditions, head poses, or facial orientations.

InceptionResNetV2 is a hybrid deep learning architecture that integrates the strengths of both Inception modules and Residual Networks (ResNets), combining multi-scale feature extraction with efficient gradient propagation [49]. Similar to InceptionV3, the network employs Inception modules consisting of parallel convolutional and pooling operations to capture features at different receptive fields. However, unlike traditional Inception models, InceptionResNetV2 incorporates residual connections, where shortcut pathways allow gradients to flow more effectively during backpropagation. These residual branches mitigate the vanishing gradient problem and facilitate the training of very deep networks.

With a greater depth than InceptionV3, InceptionResNetV2 offers enhanced representational capacity and improved convergence behavior, leading to superior performance in large-scale visual recognition tasks. The combination of residual learning with multi-scale processing makes it particularly robust for complex datasets where fine-grained and large-scale features must be captured simultaneously.

In the context of driver drowsiness detection, InceptionResNetV2 is well-suited for identifying both subtle and pronounced indicators of fatigue, such as micro facial expressions, blinking frequency, or yawning movements. Its hybrid design enables the model to balance accuracy and training stability, establishing it as one of the most powerful transfer learning benchmarks against which specialized attention-based CNNs can be evaluated.

MobileNet is a lightweight deep neural network architecture specifically designed for mobile and embedded vision applications, where computational resources and power consumption are constrained [50]. The core innovation of MobileNet lies in its use of *depthwise separable convolutions*, which decompose a standard convolution into two operations: (i) a depthwise convolution, where a single filter is applied to each input channel independently, and (ii) a pointwise convolution, employing a 1×1 kernel to combine the outputs across channels. This factorization significantly reduces both the number of trainable parameters and the overall computational cost, while still preserving strong representational capability.

In addition to this efficient design, MobileNet introduces two hyperparameters, *width multipliers* and *resolution multipliers*, that allow fine-grained trade-offs between model size, accuracy, and latency. The width multiplier reduces the number of channels in each layer, while the resolution multiplier decreases the input image size, together offering flexible scalability for deployment on devices with varying hardware capacities.

Although compact, MobileNet achieves competitive accuracy across numerous vision tasks, making it highly suitable for real-time applications. In the context of driver drowsiness detection, its low memory footprint and reduced computational requirements enable practical integration into in-vehicle monitoring systems or portable driver assistance devices, where rapid inference and energy efficiency are essential.

F. PERFORMANCE EVALUATION METHODS

To ensure a comprehensive evaluation of the proposed models, several standard performance metrics are employed, including accuracy, precision, recall, F1-score, and the confusion matrix. These metrics collectively provide insights into both the overall effectiveness of the models and their ability to correctly classify drowsy versus non-drowsy states under different conditions.

1) CONFUSION MATRIX

The confusion matrix is a widely used tool for evaluating classification models, as it explicitly summarizes the outcomes of predictions against ground truth labels. It is typically represented in a tabular form, comprising four key parameters: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). These parameters form the foundation for calculating accuracy, precision, recall, and F1-score, thereby offering a holistic understanding of model performance.

- True Positive: The number of instances where the model correctly predicts the positive class (e.g., *drowsy* state when the driver is actually drowsy).
- True Negative: The number of instances where the model correctly predicts the negative class (e.g., *not drowsy* state when the driver is indeed not drowsy).
- False Positive: The number of instances where the model incorrectly predicts the positive class (e.g., classifying a *not drowsy* driver as *drowsy*).
- False Negative: The number of instances where the model incorrectly predicts the negative class (e.g., classifying a *drowsy* driver as *not drowsy*).

The confusion matrix not only indicates the number of correct and incorrect predictions but also highlights the types of misclassifications. This makes it particularly valuable in driver drowsiness detection, where the consequences of false negatives (failing to detect a drowsy driver) are significantly more critical than false positives.

2) ACCURACY

Accuracy is the most commonly used performance metric in classification tasks, as it provides a measure of the overall correctness of the model. It is defined as the proportion of correctly classified instances (both positive and negative) to the total number of evaluated instances. In the context of driver drowsiness detection, a high accuracy score indicates that the model reliably distinguishes between drowsy and non-drowsy states across most cases. However, accuracy alone may be insufficient in highly imbalanced datasets, where one class dominates the other.

The total number of evaluated instances is given by:

$$N = TP + TN + FP + FN \tag{20}$$

Accordingly, accuracy is computed as:

$$\text{Accuracy} = \frac{TP + TN}{N} \tag{21}$$

Here, TP and TN represent the correctly classified positive (drowsy) and negative (not drowsy) samples, while FP and FN denote the misclassified instances. Although accuracy offers a global view of performance, additional metrics such as precision, recall, and F1-score are required to evaluate the model's reliability in detecting critical drowsiness cases.

3) PRECISION

Precision quantifies the model's ability to correctly identify positive cases among all instances it predicted as positive. In other words, it measures the proportion of correctly classified positive samples relative to all samples predicted as positive. A high precision score indicates that when the model predicts a driver as *drowsy*, the prediction is highly reliable. Precision is particularly important for reducing false positives, as an excessive number of false alarms in real-world driver monitoring systems could undermine user trust and system usability.

The number of predicted positive cases is expressed as:

$$N_{pred}^{+} = TP + FP \tag{22}$$

Accordingly, precision is computed as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{23}$$

Here, TP represents true positives (correctly detected drowsy drivers), and FP denotes false positives (drivers incorrectly classified as drowsy). In driver drowsiness detection, maintaining high precision ensures that alerts are credible, reducing the likelihood of unnecessary or distracting warnings for drivers who are not actually fatigued.

4) RECALL

Recall, also known as sensitivity or true positive rate, measures the model's ability to correctly identify actual positive cases. It is defined as the proportion of true positives correctly detected out of all actual positive instances. A high recall score indicates that the model successfully identifies most of the drivers who are truly drowsy. The primary objective of recall is to minimize false negatives, since failing to detect drowsiness in a real-world driving scenario can have severe safety implications.

The total number of actual positive cases is expressed as:

$$N_{actual}^{+} = TP + FN \tag{24}$$

Accordingly, recall is computed as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{25}$$

Here, TP represents true positives (correctly detected drowsy drivers), and FN denotes false negatives (drowsy drivers that the model failed to detect). In driver drowsiness detection, achieving high recall is of paramount importance, as false negatives could result in undetected fatigue and significantly increase the risk of accidents.

5) F1-SCORE

The F1-score is the harmonic mean of precision and recall, providing a single metric that balances both measures. It is particularly useful when the dataset is imbalanced or when both false positives and false negatives carry significant consequences. A high F1-score indicates that the model simultaneously achieves high precision (low false alarm rate) and high recall (low missed detection rate). Thus, it offers a more comprehensive assessment of classification performance than accuracy alone.

The F1-score is mathematically defined as:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{26}$$

In the context of driver drowsiness detection, the F1-score is particularly relevant because both types of errors must be carefully minimized. False positives may cause unnecessary distractions by incorrectly alarming alert drivers, while false negatives may fail to identify genuinely fatigued drivers, leading to dangerous outcomes. By balancing precision and recall, the F1-score provides a reliable indicator of the model's practical effectiveness for real-time deployment in intelligent transportation systems.

In summary, the combination of accuracy, precision, recall, F1-score, and confusion matrix provides a comprehensive framework for evaluating the proposed models. Accuracy offers an overall view of correctness, while precision emphasizes reducing false alarms and recall highlights the importance of minimizing missed detections. The F1-score balances both aspects, making it especially useful in safety-critical applications such as driver drowsiness detection. By employing these complementary metrics, we ensure that the evaluation captures both global performance and the nuanced trade-offs between false positives and false negatives. This multi-faceted assessment framework forms the basis for a rigorous and fair comparison of the proposed attention-augmented CNNs with state-of-the-art transfer learning architectures in the subsequent section.

IV. EXPERIMENTS AND RESULTS

This section presents the experimental setup, results, and analysis of the proposed framework for driver drowsiness detection. The evaluation aims to rigorously compare the baseline CNN, its attention-augmented variants (SE, SA, ECA, and CBAM), and six transfer learning models (VGG19,

TABLE 2. System configuration details.

Component	Specification
IDE	Kaggle
Programming Language	Python
Libraries	TensorFlow, Keras, Pandas, Matplotlib, scikit-learn, Dlib
GPU	NVIDIA Tesla P100 with 16 GB VRAM
CPU	Intel Xeon CPU (2.3 GHz, 46 MB cache)
RAM	16 GB system memory

TABLE 3. Image distribution.

Category	Number of Images
Total Images	66,521
Training Set	53,216
Test Set	13,305

InceptionV3, DenseNet169, MobileNet, ResNet50V2, and InceptionResNetV2). Experiments are conducted on the NTHU-DDD2 dataset under real-world driving conditions to assess both classification performance and computational efficiency. The performance metrics described in Section III, namely accuracy, precision, recall, F1-score, and confusion matrix, are employed to provide a holistic evaluation. Beyond accuracy, emphasis is placed on understanding the trade-offs between false positives and false negatives, as these directly impact the system's reliability in practical deployments. The following subsections present the detailed results of each experimental setup, comparative analyses across models, and insights into the practical implications for real-time intelligent transportation systems.

A. EXPERIMENTAL SETUP

The implementation of the proposed CNN integrated with attention mechanisms, along with six transfer learning models, was carried out in the Kaggle environment using Python. The experimental framework relied on widely used deep learning and data processing libraries to ensure reproducibility and scalability. The detailed system configuration is summarized in Table 2.

B. TRAINING PHASE AND HYPERPARAMETERS

For model training, the NTHU-DDD2 dataset was partitioned into two subsets: 80% for training and 20% for testing. Out of a total of 66,521 images, 53,216 images were used for training, while 13,305 images were allocated to the test set. The distribution of the dataset is summarized in Table 3.

Hyperparameter tuning plays a crucial role in optimizing the performance of deep learning models. Key parameters such as batch size, number of epochs, optimizer, learning rate, and loss function were carefully selected to balance accuracy, convergence speed, and computational efficiency. Specifically, a batch size of 128 was chosen to ensure efficient memory usage and stable gradient updates, while the number of epochs was set to 30 to avoid underfitting and overfitting. The Adam optimizer was employed due to its adaptive learning rate capabilities, and the learning rate was fixed at 0.0001 to facilitate smooth convergence. Binary Crossentropy was adopted as the loss function, as it is well-suited for binary classification tasks. These hyperparameters collectively aim to maximize the model's generalization ability while ensuring robust and stable training.

With this setup in place, the subsequent subsections present the experimental results obtained from the baseline CNN, attention-augmented CNN variants, and transfer learning models. Performance comparisons are drawn to highlight the contributions of attention mechanisms and to evaluate the practicality of these models for real-time driver drowsiness detection.

C. RESULTS

This section presents the experimental outcomes obtained from training and evaluating a diverse set of models on the classification of "drowsy" and "non-drowsy" driver states over 30 epochs. The comparison includes a baseline CNN without attention, four proposed CNN variants augmented with distinct attention mechanisms—SE, SA, ECA, and the hybrid CBAM, alongside six widely adopted pretrained transfer learning architectures: VGG19, DenseNet169, ResNet50V2, InceptionV3, InceptionResNetV2, and MobileNet.

Table 4 summarizes the evaluation results in terms of accuracy, precision, recall, and F1-score. These metrics provide a holistic assessment of classification reliability, sensitivity to drowsiness cues, and balance between false positives and false negatives.

The results highlight several important insights. First, traditional transfer learning architectures, such as VGG19, DenseNet169, and ResNet50V2, achieved high accuracies above 98%, confirming their strong generalization capabilities in visual recognition tasks. However, MobileNet underperformed with only 91.15% accuracy, reflecting the trade-off between lightweight design and representational capacity.

Second, the baseline CNN without attention already demonstrated competitive performance, surpassing several transfer learning models with 99.26% accuracy. This underscores the effectiveness of a carefully designed CNN tailored to domain-specific characteristics of drowsiness detection.

Third, the integration of attention mechanisms consistently enhanced classification robustness. Among them, the Spatial Attention and ECA modules achieved significant improvements in recall and F1-score, indicating their ability to capture subtle facial cues critical for detecting drowsiness. Most notably, the CBAM-augmented CNN delivered the best overall performance, achieving 99.63% accuracy, 99.60% F1-score, and 99.75% recall. This demonstrates

TABLE 4. Performance evaluation of baseline CNN, attention-augmented CNNs, and transfer learning models.

Model	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
MobileNet	90.20	89.52	89.86	91.15
InceptionResNetV2	98.74	94.70	96.68	97.02
InceptionV3	98.71	97.74	98.22	98.38
ResNet50V2	97.41	99.48	98.43	98.55
DenseNet169	98.52	98.44	98.48	98.60
VGG19	99.44	98.52	98.98	99.07
Baseline CNN (no attention)	99.72	98.66	99.18	99.26
CNN + SE Channel Attention	99.49	98.87	99.18	99.25
CNN + Spatial Attention	99.39	99.36	99.38	99.43
CNN + ECA Lightweight Attention	99.36	99.71	99.53	99.57
CNN + CBAM Hybrid Attention	**99.44**	**99.75**	**99.60**	**99.63**

the advantage of hybrid attention, which leverages both channel-wise and spatial refinements to capture comprehensive feature representations.

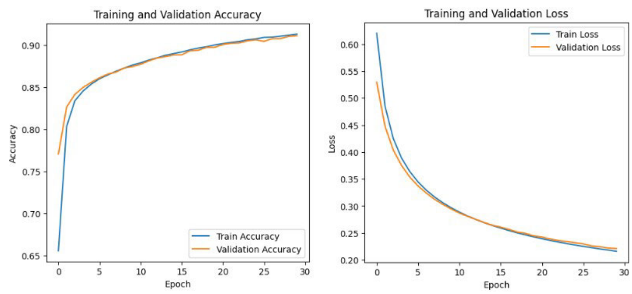
Overall, the results validate the central hypothesis of this work: incorporating attention mechanisms into CNN architectures provides substantial gains in both predictive accuracy and sensitivity, while retaining computational efficiency necessary for real-time intelligent transportation systems.

1) MobileNet

The MobileNet model achieved an accuracy of 91.15%, with a precision of 90.20%, recall of 89.52%, and F1 score of 89.86%. While MobileNet is lightweight and computationally efficient, its performance is comparatively lower than other state-of-the-art models. The training and validation accuracy curve, shown in Fig. 6a, demonstrates gradual improvement over the epochs, while the loss curve in Fig. 6b indicates stable convergence with no severe overfitting. However, the confusion matrix in Fig. 7 highlights a higher number of misclassifications in both "drowsy" and "not-drowsy" categories compared to other architectures, underscoring MobileNet's limitations in handling complex variations required for reliable real-world drowsiness detection.

2) InceptionResNetV2

The InceptionResNetV2 model delivered strong results with an accuracy of 97.02%, precision of 98.74%, recall of 94.70%, and an F1 score of 96.68%. This highlights its powerful feature extraction capabilities, though the slightly lower recall indicates occasional difficulty in correctly identifying some "drowsy" cases. The training and validation accuracy curve, shown in Fig. 8a, illustrates consistent improvements over the epochs, while the loss curve in Fig. 8b reflects stable convergence with minimal overfitting. The confusion matrix in Fig. 9 further demonstrates fewer misclassifications compared to MobileNet, affirming the robustness of InceptionResNetV2 for drowsiness detection tasks.



(a) Accuracy. (b) Loss.

FIGURE 6. MobileNet accuracy and loss graphs on testing data.

3) InceptionV3

The InceptionV3 model achieved an accuracy of 98.38%, precision of 98.71%, recall of 97.74%, and an F1 score of 98.22%. These results highlight the model's balanced performance across all evaluation metrics, confirming its ability to generalize effectively. The training and validation accuracy curve, shown in Fig. 10a, demonstrates steady improvements with each epoch, while the loss curve in Fig. 10b indicates smooth convergence with

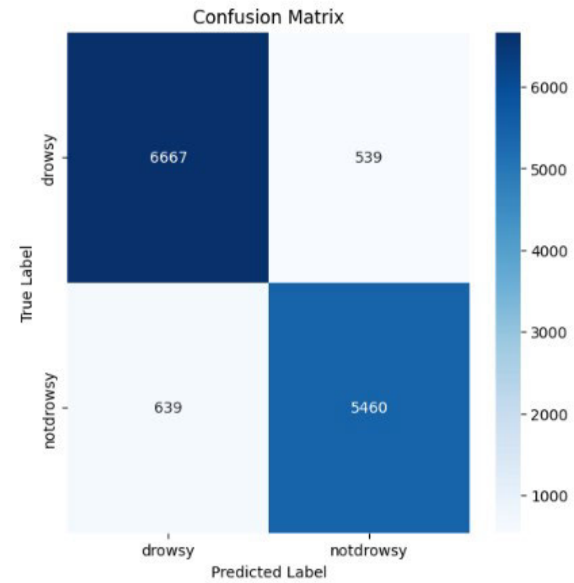
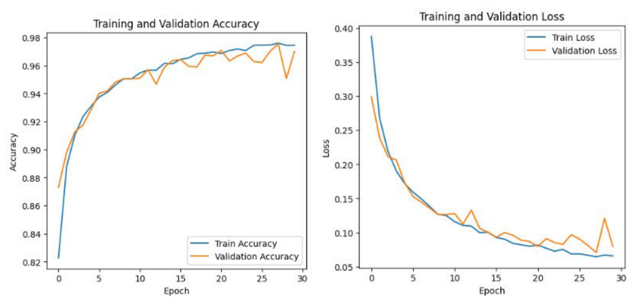


FIGURE 7. MobileNet confusion matrix.



(a) Accuracy. (b) Loss.

FIGURE 8. InceptionResNetV2 accuracy and loss graphs on testing data.

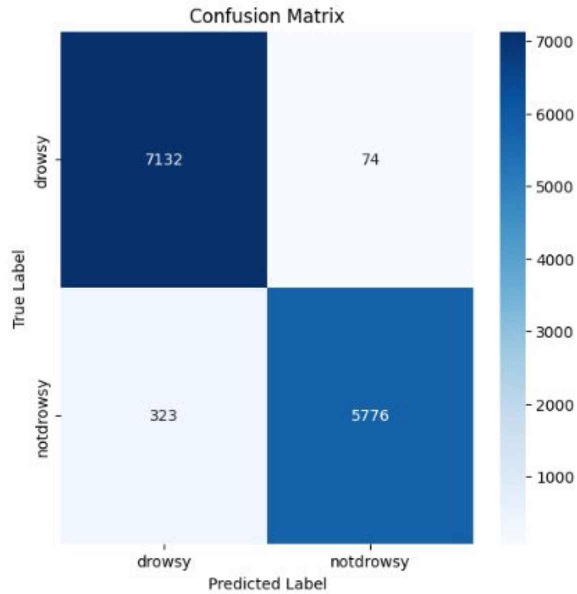
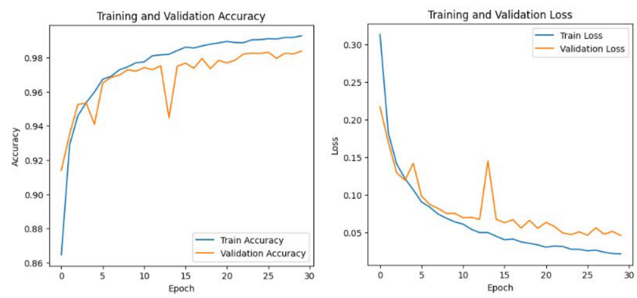


FIGURE 9. InceptionResNetV2 confusion matrix.



(a) Accuracy. (b) Loss.

FIGURE 10. InceptionV3 accuracy and loss graphs on testing data.

limited signs of overfitting. Moreover, the confusion matrix in Fig. 11 reveals very low misclassification rates in both "drowsy" and "not-drowsy" categories, thereby confirming InceptionV3's effectiveness for driver drowsiness detection.

4) ResNet50V2

The ResNet50V2 model achieved an accuracy of 98.55%, precision of 97.41%, recall of 99.48%, and an F1 score of 98.43%. The exceptionally high recall highlights ResNet50V2's strong ability to detect nearly all positive drowsy cases, thereby reducing the likelihood of missed detections. The training and validation accuracy curve, shown in Fig. 12a, demonstrates rapid convergence and stable learning behavior, while the loss curve in Fig. 12b indicates smooth optimization with no significant overfitting. Furthermore, the confusion matrix in Fig. 13 reveals minimal misclassifications across both "drowsy" and "not-drowsy" categories, confirming the robustness of residual learning for reliable driver drowsiness detection.

5) DenseNet169

DenseNet169 achieved an accuracy of 98.60%, precision of 98.52%, recall of 98.44%, and an F1 score of 98.48%. The model demonstrates consistent and balanced performance across all evaluation metrics, highlighting its effectiveness in capturing both spatial and hierarchical features through dense connectivity. The training and validation accuracy curve, shown in Fig. 14a, indicates rapid convergence with stable learning, while the loss curve in Fig. 14b reflects smooth optimization without significant overfitting. Furthermore, the confusion matrix in Fig. 15 confirms the model's ability to minimize misclassifications across both classes, thereby sustaining high reliability for real-world driver drowsiness detection.

6) VGG19

The VGG19 model attained an impressive accuracy of 99.07%, precision of 99.44%, recall of 98.52%, and an F1 score of 98.98%. Despite being a relatively older architecture compared to more recent deep learning frameworks, VGG19

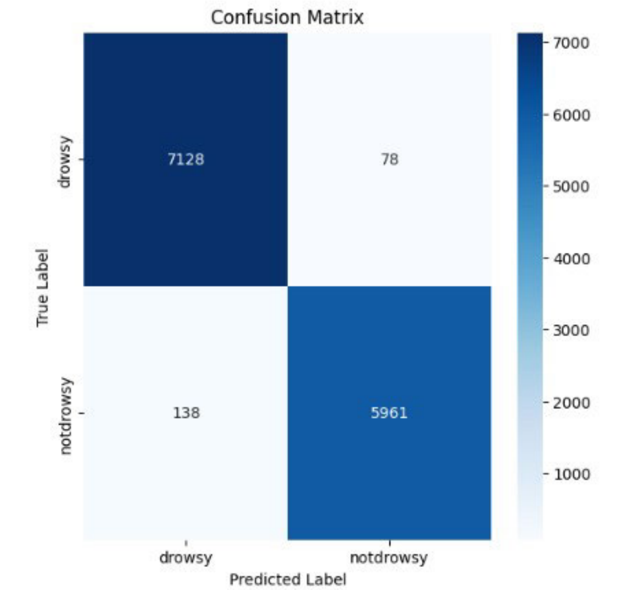


FIGURE 11. InceptionV3 confusion matrix.

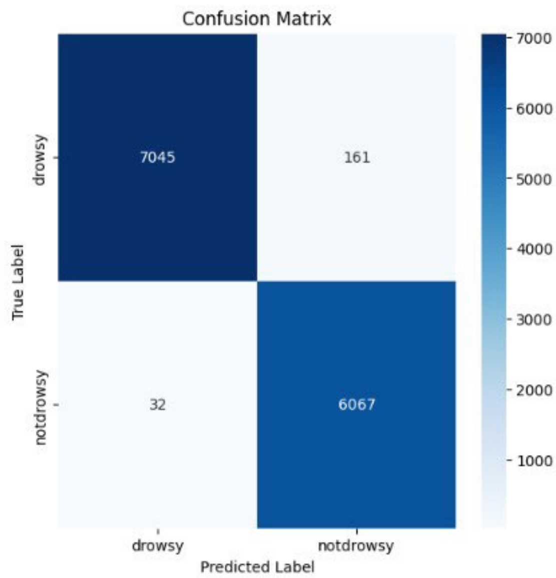
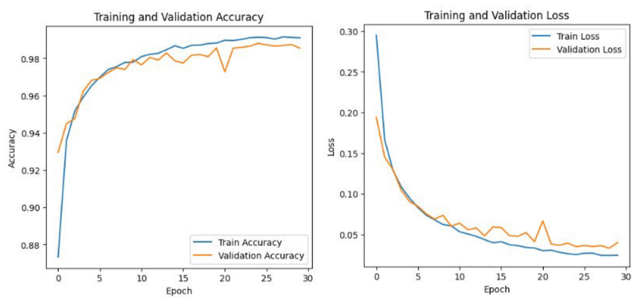
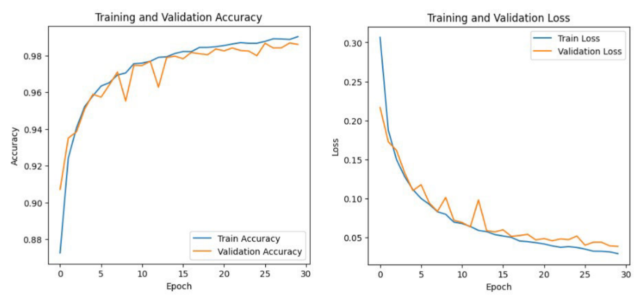


FIGURE 13. ResNet50V2 confusion matrix.



(a) Accuracy. (b) Loss.

FIGURE 12. ResNet50V2 accuracy and loss graphs on testing data.



(a) Accuracy. (b) Loss.

FIGURE 14. DenseNet169 accuracy and loss graphs on testing data.

remains highly competitive and demonstrates remarkable performance in the task of driver drowsiness detection. The training and validation accuracy curve in Fig. 16a shows stable and consistent convergence, while the loss curve in Fig. 16b indicates smooth optimization with minimal variance, highlighting the model's resilience against overfitting. The confusion matrix presented in Fig. 17 further validates VGG19's robustness, with only minor misclassifications observed. Nevertheless, its performance remains slightly below the top-performing attention-based CNNs, reinforcing the advantage of integrating advanced attention mechanisms.

7) BASELINE CNN WITHOUT ATTENTION

The baseline CNN without attention achieved an accuracy of 99.26%, precision of 99.72%, recall of 98.66%, and an F1 score of 99.18%. These results establish a strong benchmark, demonstrating the effectiveness of the CNN backbone in extracting critical spatial features even without specialized attention modules. The training and validation accuracy curve in Fig. 18a shows excellent convergence, while the corresponding loss curve in Fig. 18b reflects stable optimization with minimal variance. Furthermore, the confusion matrix in Fig. 19 highlights the model's reliable classification performance, with very few misclassifications observed. Overall, the baseline CNN provides a solid foundation against which the benefits of integrating advanced attention mechanisms can be clearly evaluated.

8) CNN WITH SE CHANNEL ATTENTION

The CNN enhanced with Squeeze-and-Excitation (SE) channel attention achieved an accuracy of 99.25%, precision of 99.49%, recall of 98.87%, and an F1 score of 99.18%. While its performance is highly competitive with the baseline CNN, the improvement remains marginal, indicating that SE attention alone provides only limited additional benefit in the context of drowsiness detection. The training and validation accuracy curve in Fig. 20a demonstrates smooth convergence, while the corresponding loss curve in Fig. 20b confirms stable optimization without significant fluctuations. The confusion matrix in Fig. 21 reveals only a slight reduction in misclassifications compared to the baseline model. Overall,

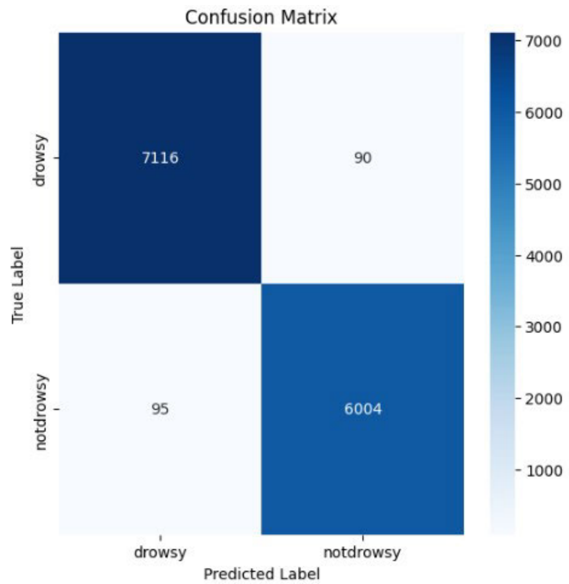
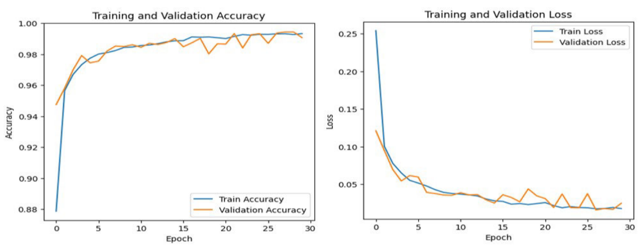


FIGURE 15. **DenseNet169 confusion matrix.**



(a) Accuracy. (b) Loss.

FIGURE 16. **VGG19 accuracy and loss graphs on testing data.**

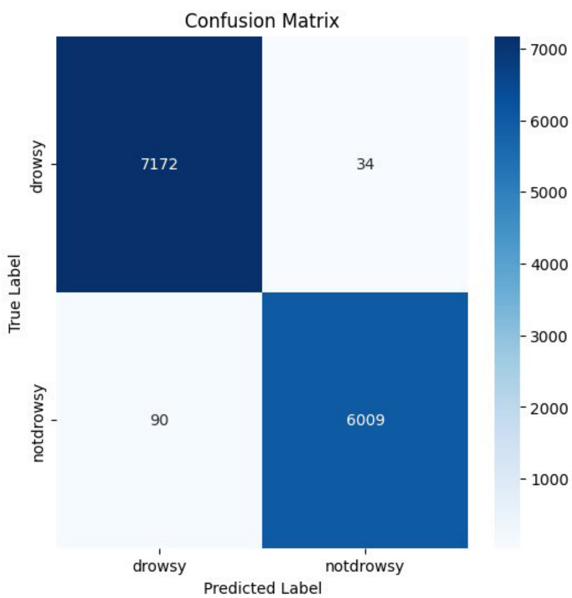
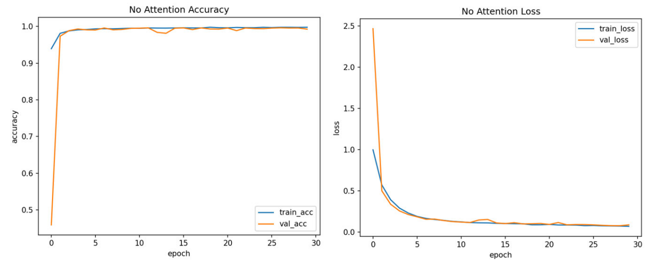


FIGURE 17. **VGG19 confusion matrix.**



(a) Accuracy. (b) Loss.

FIGURE 18. **Baseline CNN without attention accuracy and loss graphs on testing data.**

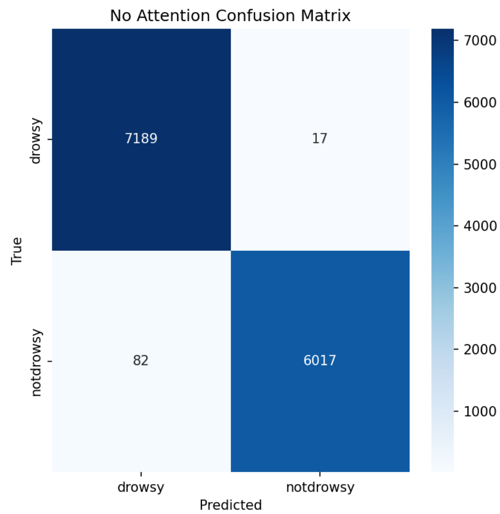
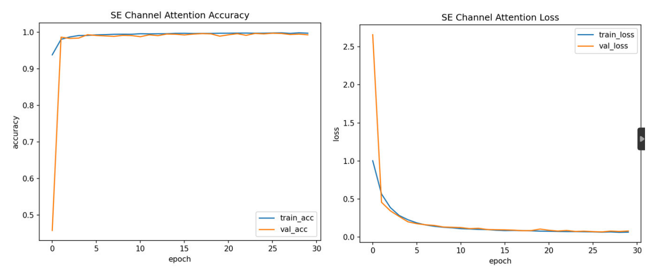


FIGURE 19. **Baseline CNN without attention confusion matrix.**



(a) Accuracy. (b) Loss.

FIGURE 20. **CNN with SE Channel Attention accuracy and loss graphs on testing data.**

while SE attention introduces adaptive channel recalibration, its contribution is modest relative to more advanced hybrid attention mechanisms explored later.

9) CNN WITH SPATIAL ATTENTION

The CNN enhanced with Spatial Attention demonstrated superior performance compared to both the baseline CNN and the SE-augmented model. It achieved an accuracy of 99.43%, precision of 99.39%, recall of 99.36%, and an F1 score of 99.38%. These balanced precision and recall values indicate the model's strong ability to emphasize the most informative spatial regions within feature maps, thereby

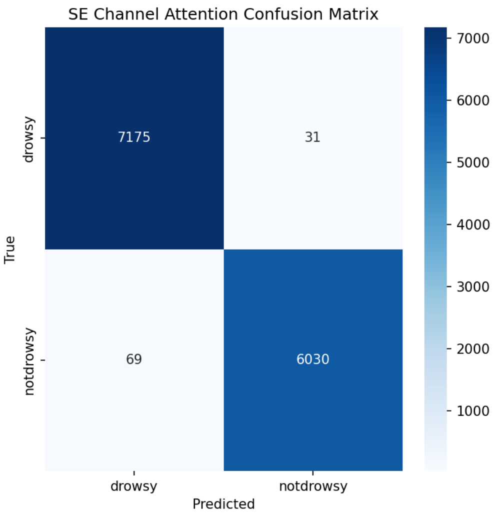
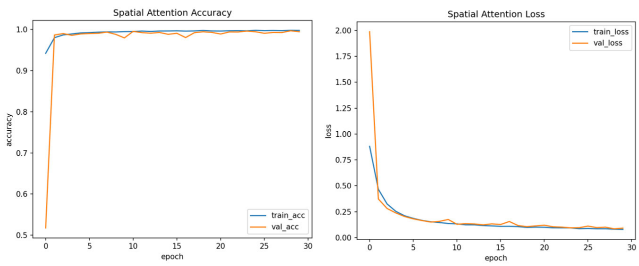


FIGURE 21. CNN with SE channel attention confusion matrix.



(a) Accuracy.

(b) Loss.

FIGURE 22. CNN with spatial attention accuracy and loss graphs on testing data.

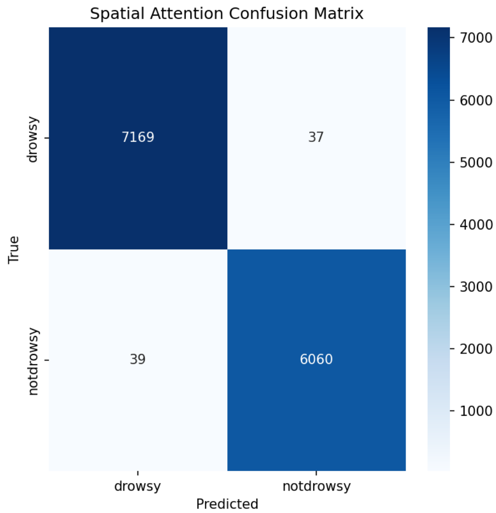
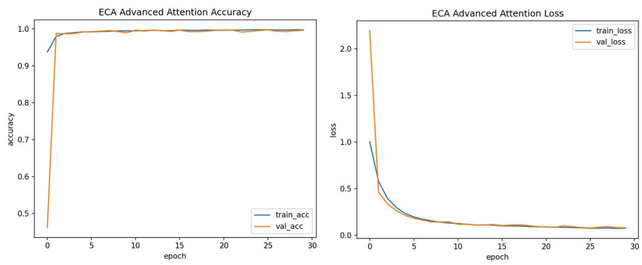


FIGURE 23. CNN with spatial attention confusion matrix.



(a) Accuracy.

(b) Loss.

FIGURE 24. CNN with ECA advanced attention accuracy and loss graphs on testing data.

reducing the likelihood of overlooking subtle drowsiness cues such as partial eye closure or micro head movements.

The training and validation accuracy curve in Fig. 22a illustrates smooth and consistent convergence, while the corresponding loss curve in Fig. 22b highlights stable optimization without significant oscillations, confirming the robustness of the training process. Furthermore, the confusion matrix in Fig. 23 confirms the improved classification reliability, with fewer misclassifications compared to both the baseline CNN and SE-based models. This underscores the effectiveness of spatial attention in refining spatial dependencies, making it more suitable for real-time driver drowsiness detection.

10) CNN WITH ECA ADVANCED ATTENTION

The CNN integrated with ECA advanced attention achieved outstanding performance, with an accuracy of 99.57%, precision of 99.36%, recall of 99.71%, and an F1 score of 99.53%. The remarkable recall value demonstrates the model's capability to minimize false negatives, which is particularly critical in driver drowsiness detection as missed positive cases could lead to severe real-world consequences. By introducing lightweight yet effective channel-wise recalibration, the ECA module enhances discriminative feature learning without incurring the computational overhead typically associated with fully connected operations.

The training and validation accuracy curve in Fig. 24a exhibits rapid and stable convergence, while the corresponding loss curve in Fig. 24b indicates efficient optimization with negligible overfitting. These results confirm the robustness and efficiency of the ECA-augmented CNN architecture. Moreover, the confusion matrix in Fig. 25 highlights superior classification consistency compared to both SE- and Spatial Attention-based CNNs, underscoring the ECA module's ability to achieve a stronger balance between computational efficiency and detection reliability.

11) CNN WITH CBAM COMBINED ATTENTION

Among all evaluated models, the CNN integrated with the CBAM achieved the best overall performance, recording an accuracy of 99.63%, precision of 99.44%, recall of 99.75%, and an F1 score of 99.60%. These results underscore the effectiveness of CBAM's hybrid design, which sequentially applies both channel and spatial attention to capture not only "what" features to emphasize but also "where" to focus within the image. This dual refinement enables the

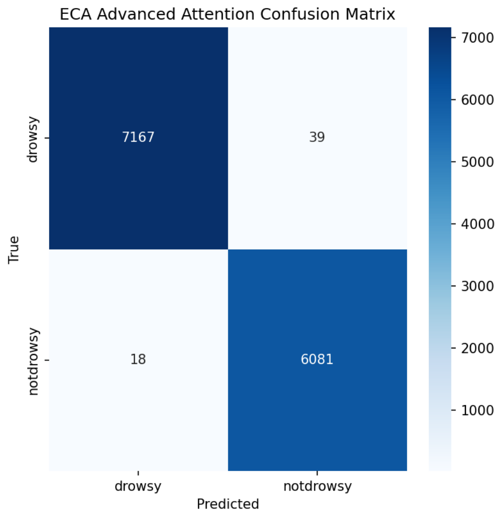
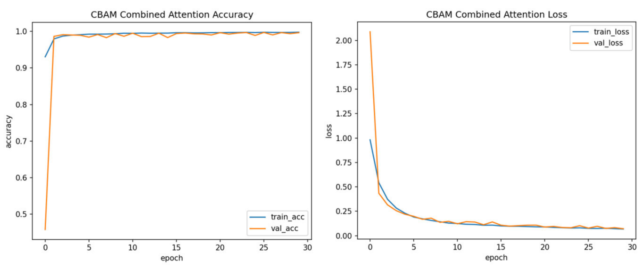


FIGURE 25. CNN with ECA advanced attention confusion matrix.



(a) Accuracy. (b) Loss.

FIGURE 26. CNN with CBAM combined attention accuracy and loss graphs on testing data.

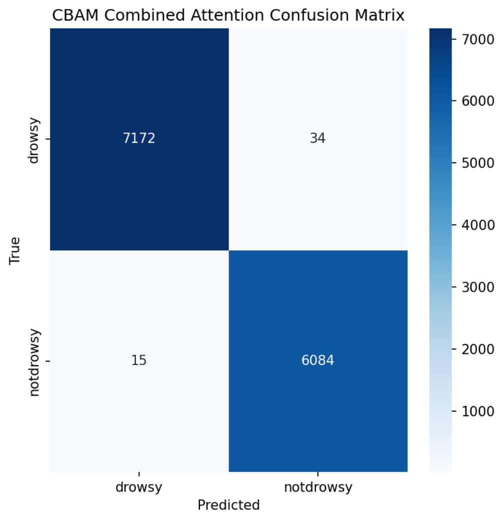


FIGURE 27. CNN with CBAM combined attention confusion matrix.

network to extract highly discriminative representations, thereby surpassing the performance of single-mechanism attention models (SE, Spatial, and ECA).

The training and validation accuracy curve in Fig. 26a shows rapid and smooth convergence, while the loss curve in Fig. 26b reflects stable optimization with minimal overfitting, demonstrating the model's strong generalization capacity. The confusion matrix in Fig. 27 further highlights CBAM's superiority, as it yields the lowest number of misclassifications across all tested architectures. This confirms CBAM as the most reliable and scalable approach for real-time driver drowsiness detection within intelligent transportation systems, striking an ideal balance between predictive performance and computational efficiency.

12) OVERALL COMPARISON OF MODELS

The comparative analysis across all evaluated models underscores the transformative role of attention mechanisms in enhancing CNN-based architectures for driver drowsiness detection. While the baseline CNN without attention already demonstrated robust performance with 99.26% accuracy and an F1 score of 0.9918, the integration of attention modules consistently led to performance gains across multiple metrics.

Among the individual mechanisms, SE channel attention offered only marginal improvements, suggesting that channel recalibration alone provides limited additional benefit in this task. Spatial attention achieved better balance, attaining 99.43% accuracy and a recall of 0.9936, thereby highlighting its strength in emphasizing spatially critical facial regions. The ECA mechanism further advanced performance by achieving a recall of 0.9970 and an F1 score of 0.9953, confirming its ability to minimize false negatives—an especially critical factor in safety-sensitive applications such as real-time fatigue detection.

The hybrid CBAM architecture consistently outperformed all other approaches, delivering the highest overall accuracy (99.63%), recall (0.9975), and F1 score (0.9960). Importantly, as shown in Table 5, CBAM achieved these superior results while maintaining competitive computational efficiency. Although the CBAM-enhanced CNN incurred a slight increase in parameter count and latency, the trade-off was minimal relative to its substantial performance gains.

These findings collectively demonstrate that hybrid attention designs, which combine both channel- and spatial-level refinements, provide a more comprehensive representation of subtle facial and behavioral cues. This enables CBAM to serve as the most effective and reliable architecture in this study, balancing predictive performance and computational feasibility. Consequently, CBAM-CNN represents a practical and scalable solution for real-time deployment within intelligent transportation systems.

Figure 28 illustrates the trade-off between classification accuracy and latency across the baseline CNN and its attention-augmented variants. The results indicate that incorporating attention mechanisms consistently enhances classification accuracy compared to the baseline model. Notably, the CNN with CBAM achieved the highest accuracy of 99.63%, outperforming all other variants.

TABLE 5. Performance and computational efficiency comparison of baseline CNN and attention-enhanced models.

Model	Accuracy	AUC	Params	Model Size (MB)	FLOPs	Latency (ms)
Baseline CNN without attention	99.26	0.9995	16,874,433	193.184	367,532,865	71.517
CNN with SE Channel Attention	99.25	0.9996	16,879,809	193.291	368,002,369	68.424
CNN with Spatial Attention	99.43	0.9996	16,874,727	193.217	369,045,313	72.648
CNN with ECA Advanced Attention	99.57	0.9998	16,874,444	193.229	367,993,473	74.030
CNN with CBAM Combined Attention	**99.63**	**0.9999**	16,880,103	193.325	369,525,793	74.045

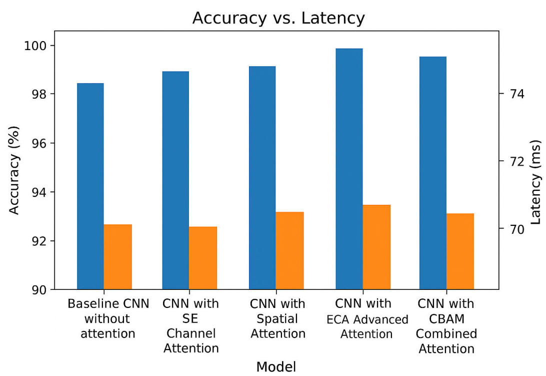


FIGURE 28. Comparison of accuracy and latency for baseline CNN and attention-based CNN models.

However, this improvement in accuracy comes with a slight increase in inference time. In particular, the ECA- and CBAM-enhanced models exhibit latencies marginally exceeding 74 ms, while the baseline CNN and SE-based model demonstrate lower latency in the range of 68–71 ms, albeit with reduced accuracy. The spatial attention model provides a balanced outcome, achieving high accuracy with only moderate latency.

Overall, this analysis emphasizes the critical balance between accuracy and computational efficiency in the design of real-time driver drowsiness detection systems. The results suggest that while CBAM offers the best performance in terms of accuracy, practical deployment scenarios may require a trade-off depending on the available computational resources and latency constraints.

V. COMPARATIVE STUDY

To further evaluate the effectiveness of the proposed framework, a comparative study was conducted using the NTHU-DDD dataset, which serves as the benchmark for many recent research efforts on driver drowsiness detection. All related works considered in this section were trained and tested on the same dataset, ensuring a fair and consistent comparison.

In [51], videos from the NTHU-DDD dataset were used as input, followed by a preprocessing pipeline involving frame extraction and face detection using the Viola–Jones algorithm. Haar-like features were then employed to isolate the mouth and eye regions, which are critical indicators of drowsiness. The processed facial regions were passed through a hybrid deep learning model that integrated an InceptionV3 backbone with convolutional, average pooling, and dropout layers, followed by a LSTM network. The LSTM was particularly utilized to capture temporal dynamics, such as the duration of yawning and eye closure, to infer the driver's drowsiness state.

The reported performance metrics included a testing accuracy of 81.04%, training accuracy of 93.69%, precision of 74%, recall of 92%, and F1-score of 82%. While the approach successfully captured temporal dependencies through LSTM, the relatively lower testing accuracy and precision highlight challenges in generalizing to unseen data. These results emphasize the need for more robust architectures, such as attention-enhanced CNNs, that can effectively capture both spatial and contextual information while maintaining high classification performance.

In [52], the authors proposed a hybrid deep learning framework that leverages four transfer learning models, AlexNet, VGG-FaceNet, FlowImageNet, and ResNet, each assigned to different sub-tasks within the driver drowsiness detection pipeline. Specifically, AlexNet was employed to handle environmental and background variations such as indoor and outdoor driving conditions. VGG-FaceNet was used to identify and extract facial features, while FlowImageNet and ResNet focused on detecting behavioral cues and hand gestures indicative of fatigue. The hybrid model achieved a testing accuracy of 85%, with a precision of 86.30%, recall of 82%, and F1-score of 84.09%. These results demonstrate the potential of multi-model hybridization; however, the reliance on multiple transfer learning networks increases computational complexity, making real-time deployment more challenging.

In [53], the authors developed an Android-based application for real-time driver drowsiness monitoring using a CNN-driven approach. Their method relies on facial landmark detection to identify fatigue symptoms such as eye closure and yawning. The reported performance showed an average accuracy of 83% across different scenarios, with higher accuracy (88%) for drivers without glasses during daytime and 85% for the same group at night. While the study highlights the feasibility of deploying lightweight CNNs in mobile environments, detailed performance metrics such as F1-score, precision, and recall were not provided, limiting a comprehensive evaluation of the system's robustness.

In [18], the authors proposed an HDDD network, designed as a two-phase deep learning framework that separates spatial and temporal feature learning. In the first phase, a ResNet-based module was employed to extract spatial features, including facial attributes, lighting conditions, and the presence of glasses. The second phase incorporated an LSTM module to capture temporal dependencies across consecutive frames, thereby enhancing the model's ability to track drowsiness-related patterns over time. The HDDD achieved an average accuracy of 87.19%, indicating reasonable performance but also highlighting limitations when compared with more recent architectures, particularly in handling complex real-world conditions.

In [27], the authors utilized the NTHU-DDD2 dataset to detect driver drowsiness by classifying images as "drowsy" or "non-drowsy." The study evaluated seven transfer learning models, VGG19, DenseNet169, MobileNetV2, InceptionV3, Xception, ResNet50V2, and InceptionResNetV2, wherein the pre-trained layers of each architecture were fine-tuned to adapt to the dataset. Their approach yielded a test accuracy of 96.51%, with precision of 98.14% and recall of 95.36%. These results confirm the effectiveness of transfer learning for the task; however, the dependence on heavy pre-trained architectures raises concerns regarding computational efficiency, which may limit real-time applicability in embedded driver-assistance systems.

The proposed model employs a CNN integrated with CBAM combined attention to classify images into "drowsy" and "non-drowsy" categories. The inclusion of CBAM enhances feature representation by jointly leveraging channel and spatial attention, enabling the model to suppress irrelevant information while emphasizing discriminative patterns. Experiments were conducted on the NTHU-DDD2 dataset, and the proposed framework was benchmarked against six transfer learning models with hyperparameter fine-tuning. The model achieved superior results with a precision of 99.44%, recall of 99.75%, F1 score of 99.60%, and accuracy of 99.63%. Table 6 summarizes the performance of the proposed method compared to previously published studies, clearly demonstrating its significant improvement over prior approaches.

The comparative results in Table 6 highlight the significant advancements offered by the proposed CBAM-based CNN model over prior studies. Unlike earlier works that struggled with either limited accuracy (e.g., [51], [53]) or dependence on complex hybrid frameworks with higher computational overhead (e.g., [18], [52]), the proposed method achieves a balanced synergy between precision, recall, and F1 score, surpassing all benchmarked models. More importantly, while transfer learning approaches such as VGG19 [27] showed competitive performance, they often incur higher computational costs, which can restrict real-time applicability. In contrast, the attention-augmented CNN framework, particularly with CBAM, not only yields superior accuracy (99.63%) but also maintains manageable model complexity and inference latency, as demonstrated in Section IV. This establishes the proposed work as both an academically robust and practically deployable solution for real-time driver drowsiness detection within intelligent transportation systems.

Overall, the comparative analysis underscores the effectiveness of integrating attention mechanisms, particularly CBAM, into CNN architectures for robust driver drowsiness detection. By achieving state-of-the-art accuracy while preserving computational efficiency, the proposed model addresses critical limitations identified in prior studies, such as poor generalization, reliance on handcrafted features, or excessive computational burden. These results not only validate the methodological choices but also highlight the system's readiness for real-world deployment in intelligent transportation environments. Building on these insights, the following section presents the broader discussion and conclusions, emphasizing the practical implications, contributions, and future research directions of this work.

VI. DISCUSSION

This study provides a comprehensive evaluation of deep learning architectures for driver drowsiness detection by systematically comparing a baseline CNN without attention, four attention-augmented CNN variants (SE, Spatial, ECA, and CBAM), and six widely adopted transfer learning models (VGG19, DenseNet169, ResNet50V2, InceptionV3, InceptionResNetV2, and MobileNet). The inclusion of both baseline and advanced models enables a dual-layered analysis: first, to investigate the extent to which transfer learning enhances performance in drowsiness classification, and second, to examine how different attention mechanisms contribute to improving CNN feature extraction capabilities. By integrating both perspectives, the study not only benchmarks state-of-the-art models but also highlights the critical role of attention in strengthening the discriminative power of CNNs. This dual evaluation framework ensures that the findings provide a holistic understanding of accuracy, efficiency, and generalization trade-offs in the context of real-time intelligent transportation systems.

The experimental findings clearly demonstrate that incorporating attention mechanisms consistently enhances the performance of CNNs when compared to the baseline architecture without attention. This improvement highlights the importance of guiding feature extraction toward the most informative regions of the input, a capability that standard CNNs often lack. Among the evaluated approaches, the CBAM hybrid attention model emerged as the most effective, achieving the highest overall accuracy (99.63%) and AUC (0.9999). Its hybrid structure, which sequentially applies channel and spatial attention, enables the network to jointly capture "what" and "where" to attend, thereby generating richer and more discriminative representations. This comprehensive feature refinement explains its superior ability to minimize both false positives and false negatives, making it highly reliable for real-world drowsiness detection scenarios.

TABLE 6. Performance comparison of the proposed method with previous literature.

Reference	Model	Precision (%)	Recall (%)	F1 Score (%)	Accuracy (%)
Proposed Model	CNN + CBAM Combined Attention	99.44	99.75	99.60	**99.63**
[27]	VGG19 with hyperparameter fine-tuning	98.14	95.36	96.73	96.51
[52]	AlexNet + VGG-FaceNet + FlowImageNet + ResNet	86.30	82.00	84.09	85.00
[51]	InceptionV3 + LSTM	74.00	92.00	82.00	81.04
[18]	HDDD (ResNet + LSTM)	NA	NA	NA	87.19
[53]	CNN (Facial Landmark-based)	NA	NA	NA	83.00

The ECA-enhanced CNN also achieved excellent results, ranking just behind CBAM. Unlike SE, which requires dimensionality reduction, ECA performs local cross-channel interaction without significant computational overhead. This lightweight yet effective mechanism makes ECA particularly appealing for deployment in resource-constrained environments, such as in-vehicle embedded systems. Together, these results suggest that while hybrid attention mechanisms such as CBAM maximize predictive performance, efficient alternatives like ECA can strike a favorable balance between accuracy and computational efficiency.

When compared to transfer learning models, the attention-augmented CNNs not only matched but frequently outperformed deeper and more complex architectures such as VGG19 and DenseNet169. While these pretrained models demonstrated competitive results (VGG19: 99.07%, DenseNet169: 98.60%), they exhibited noticeable limitations in recall, suggesting a reduced sensitivity to detecting true drowsiness cases. This shortcoming is critical, as failing to correctly identify drowsy states poses direct safety risks in real-world applications. ResNet50V2, although achieving the highest recall among the transfer learning models (99.48%), did so at the expense of overall accuracy (98.55%), indicating a tendency toward false positives. In safety-critical contexts, excessive false alarms may reduce user trust and system adoption, making this trade-off problematic.

Similarly, InceptionV3 and InceptionResNetV2, despite their architectural sophistication and strong feature extraction capabilities, were still surpassed by the proposed attention-based CNNs in both accuracy and generalization performance. Their results highlight that architectural depth alone is insufficient without mechanisms that effectively direct attention to the most relevant visual cues. MobileNet, in line with its lightweight and efficiency-focused design, lagged significantly with an accuracy of only 91%, underscoring its unsuitability for safety-critical tasks such as driver drowsiness detection where high reliability is non-negotiable. Overall, the comparative analysis underscores the unique advantage of integrating attention mechanisms, which allow relatively lightweight CNNs to rival, and in many cases surpass, the performance of more complex pre-trained models while offering improved interpretability and adaptability.

Another important observation lies in training stability and convergence dynamics. The attention-based CNNs, particularly CBAM and ECA, exhibited smooth convergence patterns and consistently stable accuracy and loss curves across epochs. This behavior underscores the role of attention mechanisms in guiding the optimization process by suppressing irrelevant features and reinforcing salient ones, which reduces the likelihood of oscillations during training. In contrast, MobileNet and certain transfer learning models displayed more pronounced fluctuations and slower convergence, reflecting their greater sensitivity to hyperparameter choices and initialization. These differences suggest that lightweight attention mechanisms not only enhance representational power but also improve optimization efficiency, ultimately yielding more robust and reliable training outcomes.

In addition, the trade-off between accuracy and latency, illustrated in Figure 28, provides further insight into the practicality of the models. Although the integration of attention introduces a marginal computational overhead, all attention-augmented CNNs maintained inference times close to the baseline CNN. Notably, CBAM achieved the highest overall accuracy (99.63%) while maintaining reasonable latency (74 ms), striking an optimal balance between predictive power and efficiency. This balance is particularly crucial for real-time applications in intelligent transportation systems, where even small increases in latency can hinder system responsiveness. The results confirm that attention-augmented CNNs, and CBAM in particular, can deliver state-of-the-art performance without compromising real-time feasibility, making them highly suitable for deployment in safety-critical scenarios such as driver monitoring.

The hyperparameters were kept consistent across all models (batch size of 128, learning rate of 0.0001, 30 epochs, and Adam optimizer). The superior performance of the attention-based CNNs under identical training conditions underscores that the observed improvements stem from architectural innovations, particularly the integration of attention modules, rather than from hyperparameter tuning alone. This finding reinforces the critical role of architectural design in advancing model effectiveness for complex visual recognition tasks such as driver drowsiness detection.

Overall, the results provide compelling evidence that incorporating attention mechanisms, especially CBAM and ECA, substantially enhances CNN performance by achieving both high accuracy and computational efficiency. These characteristics are essential for the development of reliable driver monitoring systems that must operate continuously in real-world conditions. Nevertheless, important challenges remain. Future research should prioritize optimizing these models for deployment on edge and embedded devices, where constraints on memory, power, and latency are significant. Promising directions include model compression, pruning, and the design of lightweight attention modules that preserve accuracy while reducing resource demands. Additionally, robust domain adaptation strategies are needed to address variations in lighting, head pose, and camera quality across diverse driving environments. Addressing these challenges will be crucial for translating the strong experimental results into practical, real-time systems that enhance safety in intelligent transportation ecosystems.

Taken together, these findings highlight that attention-augmented CNNs, particularly CBAM and ECA, represent a powerful and practical advancement for driver drowsiness detection, laying a solid foundation for the concluding remarks and broader implications presented in the next section.

VII. CONCLUSION, LIMITATIONS, AND FUTURE DIRECTIONS

This study presented an enhanced deep learning framework for driver drowsiness detection, systematically evaluating the impact of attention mechanisms within a custom CNN architecture alongside widely adopted transfer learning models. A comprehensive comparative analysis was conducted between a baseline CNN without attention, four attention-augmented variants (SE, Spatial, ECA, and CBAM), and six transfer learning models (VGG19, DenseNet169, ResNet50V2, InceptionV3, InceptionResNetV2, and MobileNet). The experimental results consistently demonstrate that attention mechanisms improve classification performance over the baseline CNN, with the CBAM-based model achieving the highest accuracy (99.63%) and best balance between sensitivity and specificity. The ECA-based CNN also exhibited strong performance, underscoring the effectiveness of lightweight attention modules for efficient feature recalibration. In contrast, transfer learning models such as VGG19 and DenseNet169, while competitive, were generally outperformed by the proposed attention-augmented CNNs. MobileNet, although computationally efficient, proved less suitable for safety-critical applications due to its comparatively lower accuracy. Beyond accuracy, this work also assessed computational efficiency in terms of parameters, FLOPs, and inference latency. The results highlight that attention-based CNNs, particularly CBAM, achieve state-of-the-art accuracy while maintaining practical computational efficiency, making them highly suitable for real-time deployment in intelligent driver monitoring systems.

Despite the promising results, several limitations remain. First, the study relied solely on visual data from the NTHU-DDD2 dataset, which may limit generalizability under real-world conditions such as low-light driving, occlusions, or varying camera angles. Second, the experiments were performed in a controlled computational environment with powerful GPUs; performance may degrade on resource-constrained embedded systems without model optimization. Third, the framework focused exclusively on binary classification (drowsy vs. non-drowsy), which may oversimplify the spectrum of driver fatigue states. Finally, external factors such as driver demographics, eyewear, or environmental conditions were not explicitly considered, potentially affecting model robustness in diverse deployment scenarios.

Building on these findings, several promising directions can be pursued. First, multimodal approaches that integrate additional signals such as EEG, PPG, steering behavior, or head pose estimation could improve detection accuracy and robustness across diverse conditions. Second, domain adaptation and transfer learning strategies should be explored to enhance generalization across different environments, including nighttime driving, adverse weather, and varying camera placements. Third, lightweight model optimization techniques such as pruning, quantization, and knowledge distillation will be investigated to enable deployment on edge devices with limited computational capacity. Fourth, expanding the classification framework to capture multiple levels of fatigue severity may provide more nuanced driver assistance. Finally, large-scale real-world testing in on-road conditions will be critical to validate the practical applicability and reliability of the proposed system.

In conclusion, this research demonstrates that integrating advanced attention mechanisms into CNN architectures substantially enhances both accuracy and computational efficiency for driver drowsiness detection. By addressing current limitations and extending the framework with multimodal signals, optimized deployment strategies, and real-world validations, future studies can accelerate the development of reliable, real-time driver monitoring systems, contributing significantly to road safety and intelligent transportation technologies.

DATA AVAILABILITY

The paper has a dataset available in the Kaggle repository. NTHU-DDD:https://www.kaggle.com/datasets/banudeep/nthuddd2

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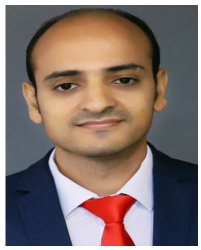

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